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Phosphorylated polypeptide antigen vaccine, preparation method therefor and application thereof

Abstract

The present invention discloses a phosphorylated polypeptide antigen vaccine, comprising at least two polypeptide fragments or conservatively modified variants thereof from human full-length Tau protein, wherein the polypeptide fragments or conservatively modified variants thereof contain phosphorylation sites. The present invention also discloses a complex vaccine formed by coupling a phosphorylated polypeptide antigen vaccine with a carrier. The polypeptide antigen vaccine and the complex vaccine can be used for preventing and/or treating tauopathy comprising Alzheimer's disease (AD).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Stage Application filed under 35 U.S.C. § 371 of International Application No. PCT/CN2019/079708, filed Mar. 26, 2019, which claims the benefit of Chinese Application No. 201810551372.1, filed May 31, 2018. Both of these applications are hereby incorporated by reference in their entireties.

REFERENCE TO A SEQUENCE LISTING

(2) A sequence listing containing SEQ ID Nos. 1-1366, created on Nov. 29, 2020, having a size of 398 kb, and entitled “108-20_sequence_listing_filed”, is provided herewith in a computer-readable nucleotide/amino acid.txt file and is specifically incorporated by reference.

TECHNICAL FIELD

(3) The present invention relates to the field of molecular biology. In particular, the present invention relates to splicing of truncated Tau proteins to form a phosphorylated polypeptide antigen vaccine. Preferably, this polypeptide antigen vaccine is connected with a suitable carrier to prepare a complex vaccine. The polypeptide antigen vaccine and the complex vaccine can be used for preventing and/or treating tauopathy comprising Alzheimer's disease (AD).

BACKGROUND ART

(4) Alzheimer's disease (AD) is a progressive neurodegenerative disease which causes deficits in cognitive function and declines in learning, memory, language, and motor functions in AD patients. With the progression of the disease, AD patients show concomitant behavioral, emotional, interpersonal, and social deterioration. Late-stage AD patients are unable to speak, understand language, and take care of themselves. Several drugs currently approved can control or relieve complications and side effects from AD and improve AD patients' life quality. Nevertheless, there is still an unmet need for treatments that directly target the disease process and have improved

therapeutic effects.

(5) AD is histologically characterized by the deposition of extraneuronal plaques, intracellular and extracellular neurofibrillary tangles and loss of neurons in the brain. The results of numerous studies demonstrate that Tau proteins play an important role in AD pathology and are also the most downstream change in the pathological changes of AD. Studies show that A β neurotoxicity in cultured neurons appears to depend on Tau proteins; and a reduction in the amount of Tau proteins in a model of tauopathy can restore memory function thereof. In addition, a reduction in endogenous Tau proteins inhibits behavioral deficits in transgenic mice that express the human amyloid precursor, without altering their A β levels. Thus, therapies targeting Tau proteins can become an effective strategy for treating tauopathy comprising AD.

(6) Tau proteins are microtubule-associated proteins. Tau proteins mediate the assembly of tubulin monomers into microtubules (MT) that constitute the neuronal microtubules network; Tau proteins carry nutrients and transmitters on microtubules through phosphorylation and dephosphorylation, which is significant for the proper formation and executive function of neuronal circuits. The binding of Tau to MT is controlled by dynamic phosphorylation and dephosphorylation, as demonstrated by experiments in vitro and in non-neuronal cells. When being hyperphosphorylated, Tau proteins will aggregate with each other and detach from the MT, and the MT will lose their stability so as to disintegrate themselves, thereby causing formation of neurofibrillary tangles (NFT) and neuronal loss.

(7) Notwithstanding AD's prevalence in human beings, its pathogenesis has not been fully elucidated so far, and the current research and treatment methods have not yet met the needs. As demonstrated by experiments conducted in a tauopathy mouse model by Asuni et al., mice vaccinated with Tau protein derived phospho-peptide showed a reduction in neurofibrillary tangles and functional improvements. Although small molecules of short peptides can often interact with immune response products, they usually cannot elicit a response alone. These peptide immunogens are also called "haptens", which cannot, on their own, produce immunogenicity or cause the production of antibodies in the body, and can only be prepared into an immunogenic composition by coupling them with a suitable carrier. On the other hand, full-length recombinant Tau proteins expressed in a prokaryotic system do not seem to be suitable as a vaccine.

(8) In view of the foregoing, there is a need in the art for the development of a method of effectively preventing and/or treating tauopathy.

SUMMARY OF THE INVENTION

(9) With respect to the aforementioned drawbacks, the present invention, based on human full-length Tau protein, designs phosphorylated polypeptide antigen vaccines, and complex vaccines which are formed by coupling the phosphorylated polypeptide antigen vaccines with suitable carriers for preventing and/or treating tauopathy comprising Alzheimer's disease (AD). The vaccine of the present invention has good safety and high immunogenicity, and can induce the production of high-titer antibodies.

(10) In particular, the first aspect of the present invention provides a phosphorylated polypeptide antigen vaccine, which comprises at least two polypeptide fragments or conservative modified variants thereof from human full-length Tau protein, wherein the polypeptide fragments or conservative modified variants thereof contain phosphorylation sites. In some embodiments, the phosphorylated polypeptide antigen vaccine contains an additional cysteine residue at its C'-terminal.

(11) In the context of the present invention, the term "phosphorylated polypeptide antigen vaccine" means a phosphorylated polypeptide, and can be used interchangeably with the term "phosphorylated polypeptide".

(12) In some embodiments, the polypeptide fragments are derived from a phosphorylation modification site-rich region of human full-length Tau protein. In some preferred embodiments, the polypeptide fragments are derived from the following regions of human full-length Tau protein:

amino acids at positions 14 to 22 of human full-length Tau protein, amino acids at positions 194 to 266 of human full-length Tau protein, and/or amino acids at positions 392 to 408 of human full-length Tau protein.

(13) In some embodiments, the polypeptide fragments are connected directly by peptide bonds or connected by amino acid linkers. In preferred embodiments, the polypeptide fragments are connected directly by peptide bonds.

(14) In some specific embodiments, the phosphorylation sites include two or more, preferably all phosphorylated amino acid sites corresponding to positions 18, 202, 205, 212, 214, 231, 235, 238, 262, 396 and 404 of the amino acid sequence of human full-length Tau protein, namely, 18(P-Tyr.sub.18), 202(P-Ser.sub.202), 205(P-Thr.sub.205), 212(P-Thr.sub.212), 214(P-Ser.sub.214), 231(P-Thr.sub.231), 235(P-Ser.sub.235), 238(P-Ser.sub.238), 262(P-Ser.sub.262), 396(P-Ser.sub.396) and 404(P-Ser.sub.404). In some preferred embodiments, 1 to 4, preferably 1 to 3, more preferably 1 to 2, and most preferably 1 Ser site of the phosphorylation sites are substituted by an aspartic acid and/or 1 to 4, preferably 1 to 3, more preferably 1 to 2, and most preferably 1 Thr site of the phosphorylation sites are substituted by a glutamic acid; the overall net charge of the polypeptide fragments or conservatively modified variants thereof thus obtained and the charge distribution on the molecules thereof remain substantially the same as those before the substitution. The simulation of phosphorylation by replacing phosphorylation sites with aspartic acids and/or glutamic acids is well known in the art.

(15) In some embodiments, the conservatively modified variant of the polypeptide fragment contained in the phosphorylated polypeptide antigen vaccine of the present invention is a variant obtained by conservatively substitution of one or more amino acids, preferably 1 to 10 amino acids, more preferably 1 to 6 amino acids, more preferably 1 to 4 amino acids, more preferably 1 to 3 amino acids and most preferably 1 amino acid of the polypeptide fragment with functionally similar amino acids. The conservatively substitution is well known in the art and includes the following 6 groups of amino acids: 1) alanine (A), serine (S) and threonine (T); 2) aspartic acid (D) and glutamic acid (E); 3) asparagine (N) and glutamine (Q); 4) arginine (R) and lysine (K); 5) isoleucine (I), leucine (L), methionine (M) and valine (V); and 6) phenylalanine (F), tyrosine (Y) and tryptophan (W).

(16) The overall net charge of the variant of the polypeptide fragment thus obtained and the charge distribution on the molecules thereof remain substantially the same as those before the substitution.

(17) In some preferred embodiments, the phosphorylated polypeptide antigen vaccine has an amino acid sequence as represented by any one of SEQ ID NOs: 1-1331, and the amino acid sequence contains two or more phosphorylation sites selected from 18(P-Tyr.sub.18), 202(P-Ser.sub.202), 205(P-Thr.sub.205), 212(P-Thr.sub.212), 214(P-Ser.sub.214), 231(P-Thr.sub.231), 235(P-Ser.sub.235), 238(P-Ser.sub.238), 262(P-Ser.sub.262), 396(P-Ser.sub.396) and 404(P-Ser.sub.404). In some embodiments, the phosphorylated polypeptide antigen vaccine has an amino acid sequence which has at least 80%, at least 85%, at least 90%, at least 95%, preferably at least 98%, more preferably at least 99% sequence identity to any one of SEQ ID NOs: 1-1331, and the phosphorylated polypeptide antigen vaccine has basically the same immunogenic activity as the original phosphorylated polypeptide antigen vaccine, and the amino acid sequence contains two or more phosphorylation sites selected from 18(P-Tyr.sub.18), 202(P-Ser.sub.202), 205(P-Thr.sub.205), 212(P-Thr.sub.212), 214(P-Ser.sub.214), 231(P-Thr.sub.231), 235(P-Ser.sub.235), 238(P-Ser.sub.238), 262(P-Ser.sub.262), 396(P-Ser.sub.396) and 404(P-Ser.sub.404).

(18) In a more preferred embodiment, the phosphorylated polypeptide antigen vaccine has an amino acid sequence as represented by any one of SEQ ID NO: 201, SEQ ID NO: 225, SEQ ID NO: 306, SEQ ID NO: 387, SEQ ID NO: 468, SEQ ID NO: 558, SEQ ID NO: 567, SEQ ID NO: 769, SEQ ID NO: 784, SEQ ID NO: 875, SEQ ID NO: 1020, SEQ ID NO: 1101, SEQ ID NO: 1182, SEQ ID NO: 1272, SEQ ID NO: 1313 and SEQ ID NO: 1330, and the amino acid sequence contains two or more phosphorylation sites selected from 18(P-Tyr.sub.18), 202(P-Ser.sub.202),

205(P-Thr.sub.205), 212(P-Thr.sub.212), 214(P-Ser.sub.214), 231(P-Thr.sub.231), 235(P-Ser.sub.235), 238(P-Ser.sub.238), 262(P-Ser.sub.262), 396(P-Ser.sub.396) and 404(P-Ser.sub.404). In some embodiments, the phosphorylated polypeptide antigen vaccine has an amino acid sequence which has at least 80%, at least 85%, at least 90%, at least 95%, preferably at least 98%, and more preferably at least 99% sequence identity to any one of the above sequences, and the phosphorylated polypeptide antigen vaccine has basically the same immunogenic activity as the original phosphorylated polypeptide antigen vaccine, and the amino acid sequence contains two or more phosphorylation sites selected from 18(P-Tyr.sub.18), 202(P-Ser.sub.202), 205(P-Thr.sub.205), 212(P-Thr.sub.212), 214(P-Ser.sub.214), 231(P-Thr.sub.231), 235(P-Ser.sub.235), 238(P-Ser.sub.238), 262(P-Ser.sub.262), 396(P-Ser.sub.396) and 404(P-Ser.sub.404).

(19) In a particularly specific embodiment, the phosphorylated polypeptide antigen vaccine has an amino acid sequence as represented by any one of SEQ ID NO: 201, SEQ ID NO: 225, SEQ ID NO: 306, SEQ ID NO: 387, SEQ ID NO: 468, SEQ ID NO: 558, SEQ ID NO: 567, SEQ ID NO: 769, SEQ ID NO: 784, SEQ ID NO: 875, SEQ ID NO: 1020, SEQ ID NO: 1101, SEQ ID NO: 1182, SEQ ID NO: 1272, SEQ ID NO: 1313 and SEQ ID NO: 1330, and the amino acid sequence respectively contains the following phosphorylation sites: 18(P-Tyr.sub.18), 202(P-Ser.sub.202), 205(P-Thr.sub.205); 18(P-Tyr.sub.18), 202(P-Ser.sub.202), 205(P-Thr.sub.205); 18(P-Tyr.sub.18), 212(P-Thr.sub.212), 214(P-Ser.sub.214); 18(P-Tyr.sub.18), 231(P-Thr.sub.231), 235(P-Ser.sub.235); 18(P-Tyr.sub.18), 238(P-Ser.sub.238), 262(P-Ser.sub.262); 18(P-Tyr.sub.18), 396(P-Ser.sub.396), 404(P-Ser.sub.404); 202(P-Ser.sub.202), 205(P-Thr.sub.205), 231(P-Thr.sub.231), 235(P-Ser.sub.235); 202(P-Ser.sub.202), 205(P-Thr.sub.205), 231(P-Thr.sub.231), 235(P-Ser.sub.235); 202(P-Ser.sub.202), 205(P-Thr.sub.205), 238(P-Ser.sub.238), 262(P-Ser.sub.262); 202(P-Ser.sub.202), 205(P-Thr.sub.205), 396(P-Ser.sub.396), 404(P-Ser.sub.404); 202(P-Ser.sub.202), 205(P-Thr.sub.205), 396(P-Ser.sub.396), 404(P-Ser.sub.404); 212(P-Thr.sub.212), 214(P-Ser.sub.214), 231(P-Thr.sub.231), 235(P-Ser.sub.235); 212(P-Thr.sub.212), 214(P-Ser.sub.214), 238(P-Ser.sub.238), 262(P-Ser.sub.262); 212(P-Thr.sub.212), 214(P-Ser.sub.214), 396(P-Ser.sub.396), 404(P-Ser.sub.404); 238(P-Ser.sub.238), 262(P-Ser.sub.262), 396(P-Ser.sub.396), 404(P-Ser.sub.404); 238(P-Ser.sub.238), 262(P-Ser.sub.262), 396(P-Ser.sub.396) and 404(P-Ser.sub.404).

(20) The second aspect of the present invention provides a complex vaccine formed by coupling the phosphorylated polypeptide antigen vaccine of the first aspect of the present invention with a carrier.

(21) In some embodiments, the carrier is selected from the group comprising human serum albumin, keyhole limpet hemocyanin, bacterium-like particles (BLP), immunoglobulin molecules, thyroglobulin, ovalbumin, bovine serum albumin component V, influenza hemagglutinin, PAN-DR binding peptide (PADRE polypeptide), malaria circumsporozoite (CS) protein, hepatitis B surface antigen (HB.sub.sAg.sub.19-28), Heat Shock Protein (HSP) 65, Bacille Calmette-Guérin (BCG), cholera toxin, attenuated cholera toxin variants, diphtheria toxin, norovirus capsid P protein, recombinant *Streptococcus* C5a peptidase, *Streptococcus pyogenes* ORF1224, *Streptococcus pyogenes* ORF1664, *Streptococcus pyogenes* ORF2452, pneumolysin, attenuated pneumolysin toxicity variants, *Chlamydia pneumoniae* ORFT367, *Chlamydia pneumoniae* ORFT858, Tetanus toxoid, HIV gp120T1, microbial surface components recognizing adhesive matrix molecules (MSCRAMMS), growth factor/hormone and chemokines, etc. Complex vaccines formed by coupling or mixing with carriers can, to the greatest extent, stimulate the body to produce a specific immune response against phosphorylated polypeptides.

(22) In some preferred embodiments, the carrier is norovirus capsid P protein.

(23) The term "norovirus capsid P protein", herein also referred to as P protein, refers to the P protein in the norovirus capsid proteins, which can self-assemble in vitro into a P particle. When used herein, P protein used at the gene level means a gene fragment, nucleotide sequence, plasmid, etc. encoding a P protein; P protein used at the protein level means a P protein monomer or

polymer.

(24) The term P particle (abbreviated as PP) refers to a protein particle formed by the in vitro self-assembly of P protein in norovirus. The most common form of P particle is a tetracosamer. When used herein, P particle (PP) is only used at the protein level and means the form of polymer (such as a tetracosamer), including various proteins used for property detection and immunization.

(25) In some specific embodiments, one amino acid in each of the three loops in loop domain of the norovirus capsid P protein is mutated to a cysteine to facilitate chemical connection with the phosphorylated polypeptide vaccine. The norovirus capsid P protein modified with this mutation was named PP-3C (whose sequence is represented by SEQ ID NO: 1357), wherein the mutation would not result in a frameshift mutation of the norovirus P protein.

(26) In other specific embodiments, a lysine is inserted into each of the three loops in loop domain of the norovirus P protein to facilitate chemical coupling with spliced phosphorylated polypeptide. The norovirus capsid P protein modified with this mutation was named PP-3K (whose sequence is represented by SEQ ID NO: 1359), wherein the mutation would not result in a frameshift mutation of the norovirus P protein.

(27) In other preferred embodiments, the carrier is bacterium-like particles (BLP).

(28) In some specific embodiments, the BLP is coupled with a phosphorylated polypeptide antigen vaccine by means of a protein adaptor (PA); specifically, the BLP is connected with a PA (a GGGGSCGGGS sequence is added to the N-terminal of the PA) (i.e., C-PA) through covalent binding to obtain C-PA-BLP, and then C-PA-BLP is coupled with a phosphorylated polypeptide antigen vaccine which contains a cysteine at the C-terminal.

(29) The term “bacterium-like particles (BLP)” refers to a new mucosal adjuvant which is obtained by hot acid treatment of *Lactococcus lactis*, and is an inanimate spherical particle with a *Lactococcus lactis* peptidoglycan shell as its main component. BLP particles, as a carrier for vaccine antigen components, can be effectively bound to the antigen and display the antigen on its surface. The peptidoglycan shell of BLP itself activates the innate immune system by interacting with Toll-like receptors to function as an adjuvant. BLP can be obtained by methods known in the art.

(30) The term “protein adaptor (PA)” refers to PA protein, which is a full-length or truncated sequence in the *Lactococcus lactis* cell wall hydrolase ACMA cell wall binding region. The PA protein used in the present invention is a product (i.e., a C-PA protein) obtained by adding a GGGGSCGGGS sequence to the N-terminal of the amino acid sequence from positions 219 to 437 of the *Lactococcus lactis* cell wall hydrolase ACMA (GenBank: U17696.1), and has a sequence as represented by SEQ ID NO: 1364.

(31) The third aspect of the present invention provides a method for preparing the complex vaccine of the second aspect of the present invention, the method comprising the following steps: 1) artificially synthesizing the phosphorylated polypeptide antigen vaccine of the first aspect of the present invention; 2) preparing a carrier to be coupled with the phosphorylated polypeptide antigen vaccine; 3) mixing the phosphorylated polypeptide antigen vaccine with the carrier to preform coupling reaction; and 4) separating and purifying the conjugate obtained in 3), thereby obtaining a complex vaccine.

(32) In some embodiments, the carrier in step 2) of the method is a norovirus capsid P protein, preferably PP-3C or PP-3K, and step 2) specifically includes i) obtaining an expression vector comprising a nucleic acid encoding a PP-3C or PP-3K protein; ii) transferring the expression vector into a recipient cell; iii) expressing the PP-3C or PP-3K protein, and allowing it to self-assemble into a recombinant P particle in the recipient cell; preferably, step 2) also comprises isolation and purification steps. In some specific embodiments, ion exchange chromatography and/or hydrophobic chromatography can be used for purification.

(33) In some embodiments, in step 3) of the method, PP-3C is used as a vaccine carrier for coupling, a preferred buffer system is an ammonium bicarbonate buffer system, and a preferred pH

ranges from 7.5 to 8.8; preferably, the phosphorylated polypeptide antigen vaccine and the carrier are mixed in a ratio of 10:1 to 100:1, and a preferred reaction temperature ranges from 2° C. to 10° C.

(34) In some embodiments, in step 3) of the method, PP-3K is used as a vaccine carrier for coupling, a preferred buffer system is a phosphate buffer system, and a preferred pH ranges from 7.0 to 8.5; preferably, the phosphorylated polypeptide antigen vaccine and the carrier are mixed in a ratio of 10:1 to 100:1, and a preferred reaction temperature ranges from 2° C. to 25° C.

(35) In other embodiments, the carrier in step 2) of the method is bacterium-like particles (BLP). In some embodiments, step 3) of the method specifically comprises i) obtaining a purified protein adaptor—C-PA protein (whose sequence is represented by SEQ ID NO: 1364); ii) connecting the carrier—bacterium-like particles (BLP)—obtained in step 2) with the C-PA protein to obtain C-PA-BLP; and iii) performing coupling reaction of C-PA-BLP with the phosphorylated polypeptide antigen vaccine; wherein a Tris buffer system is used as a buffer system, and a preferred pH ranges from 7.2 to 8.8; a preferred C-PA protein concentration is 0.1 mg/mL to 1.5 mg/mL, and a preferred reaction temperature ranges from 2° C. to 30° C. In some embodiments, the buffer system used in step 3) of the method is an ammonium bicarbonate buffer system, and a preferred pH ranges from 7.5 to 8.8; preferably, the phosphorylated polypeptide antigen vaccine and the C-PA-BLP are mixed in a ratio of 10:1 to 100:1, and a preferred reaction temperature ranges from 2° C. to 10° C.

(36) In some embodiments, step 4) of the method comprises removing carriers and polypeptide antigens which are not successfully connected by methods including centrifugation, dialysis, and ultrafiltration.

(37) The fourth aspect of the present invention provides a vaccine composition comprising the phosphorylated polypeptide antigen vaccine of the first aspect of the present invention or the complex vaccine of the second aspect of the present invention. Preferably, the vaccine composition further comprises pharmaceutically acceptable adjuvants.

(38) In some embodiments, the pharmaceutically acceptable adjuvants are selected from one or more of CpG, MF59, AS02, AS03, Freund's complete adjuvant and Freund's incomplete adjuvant.

(39) The fifth aspect of the present invention provides use of the phosphorylated polypeptide antigen vaccine of the first aspect of the present invention or the complex vaccine of the second aspect of the present invention or the vaccine composition of the fourth aspect of the present invention for preparing a medicament for prevention and/or treatment of neurodegenerative disorders.

(40) In the present invention, the “neurodegenerative disorders” include, but not limited to, AD, Creutzfeldt-Jacob Syndrome, Dementia pugilistica, Down's Syndrome, Gerstmann-Sträussler-Scheinker disease, inclusion-body myositis, prion protein cerebral amyloid angiopathy, traumatic brain injury, amyotrophic lateral sclerosis/parkinsonism syndrome-dementia syndrome, argyrophilic grain dementia, corticobasal degeneration, diffuse neurofibrillary tangles with calcification, frontotemporal dementia with parkinsonism syndrome linked to chromosome 17, Hallevorden-Spatz disease, multiple system atrophy, Niemann-Pick disease type C, Pick's disease, progressive subcortical gliosis and progressive supranuclear palsy; preferably the “neurodegenerative disorder” is AD.

(41) In some embodiments, the vaccine or vaccine composition is preferably immunized subcutaneously, intraperitoneally or intramuscularly, and more preferably immunized intramuscularly.

(42) The sixth aspect of the present invention provides use of the phosphorylated polypeptide antigen vaccine of the first aspect of the present invention or the complex vaccine of the second aspect of the present invention or the vaccine composition of the fourth aspect of the present invention for preparing a medicament for maintaining or improving, preferably recovering, and more preferably completely recovering the cognitive memory of mammals, especially human beings.

- (43) The seventh aspect of the present invention provides a method for treating or preventing neurodegenerative disorders, comprising administering to a subject the phosphorylated polypeptide antigen vaccine of the first aspect of the present invention or the complex vaccine of the second aspect of the present invention or the vaccine composition of the fourth aspect of the present invention.
- (44) In some embodiments, the subject is a mammal, preferably a human.
- (45) In some embodiments, the vaccine or vaccine composition is preferably administered subcutaneously, intraperitoneally or intramuscularly, and more preferably administered intramuscularly.
- (46) The eighth aspect of the present invention provides a method for maintaining or improving, preferably recovering, and more preferably completely recovering the cognitive memory of a subject, the method comprising administering to the subject the phosphorylated polypeptide antigen vaccine of the first aspect of the present invention or the complex vaccine of the second aspect of the present invention or the vaccine composition of the fourth aspect of the present invention.
- (47) The ninth aspect of the present invention provides a norovirus capsid P protein to be coupled with the phosphorylated polypeptide antigen vaccine of the first aspect of the present invention, wherein one amino acid in each of the three loops in loop domain of the norovirus capsid P protein is mutated to a cysteine—the P protein thus obtained is called PP-3C (with a sequence as represented by SEQ ID NO: 1357), or a lysine is inserted into each of the three loops in loop domain of the norovirus capsid P protein—the P protein thus obtained is called PP-3K (with a sequence as represented by SEQ ID NO: 1359).
- (48) The tenth aspect of the present invention provides a method for preparing the norovirus capsid P protein of the ninth aspect of the present invention, comprising the following steps: i) obtaining an expression vector comprising a nucleic acid encoding a PP-3C or PP-3K protein; ii) transferring the expression vector into a recipient cell; iii) expressing the PP-3C or PP-3K protein, and allowing it to self-assemble into a recombinant P particle in the recipient cell; preferably, the method also comprises isolation and purification steps. In some specific embodiments, ion exchange chromatography and/or hydrophobic chromatography can be used for purification.
- (49) The eleventh aspect of the present invention provides a protein adaptor with a GGGGSCGGGS sequence inserted at its N-terminal—the protein adaptor thus obtained is C-PA (whose sequence is represented by SEQ ID NO: 1364).
- (50) The twelfth aspect of the present invention provides a method for preparing the protein adaptor of the eleventh aspect of the present invention, comprising the following steps: i) obtaining an expression vector comprising a nucleic acid encoding a C-PA protein; ii) transferring the expression vector into a host cell; and iii) expressing the C-PA protein, and obtaining the C-PA protein by isolating and purifying.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIGS. 1A to 1E show the maps of the various recombinant plasmid constructs. FIG. 1A is a schematic diagram of pET26b-PP plasmid that has been constructed; FIG. 1B is a schematic diagram of pET26b-PP-3C plasmid that has been constructed; FIG. 1C is a schematic diagram of pET28a-PP plasmid that has been constructed; FIG. 1D is a schematic diagram of pET28a-PP-3K plasmid that has been constructed; FIG. 1E is a schematic diagram of pColdIV-PP-3C plasmid that has been constructed.
- (2) FIGS. 2A to 2C show the recombinant pET26b plasmid construct and the results of purification and expression. FIG. 2A is a schematic diagram of a PP-3C particle. FIG. 2B is a schematic

diagram of identification of pET26b—PP-3C plasmid that has been constructed by double-enzyme digestion; as shown in FIG. 2B, the target band is at 1,000 bp. FIG. 2C is a diagram of native gel electrophoresis of PP-3C and PP-3K protein samples; as shown in FIG. 2C, both of the protein bands appeared above 250 kDa, and it is determined from analysis that they are in the form of a dodecamer and a tetracosamer.

(3) FIGS. 3A to 3C show the recombinant pET28a plasmid construct and the results of purification and expression. FIG. 3A is a schematic diagram of a PP-3K particle. FIG. 3B is a schematic diagram of identification of pET28a-PP-3K plasmid that has been constructed by double-enzyme digestion; as shown in FIG. 3B, the target band is at 1,000 bp. FIG. 3C is a SDS-PAGE graph of the target protein elution peak protein sample; as shown in FIG. 3C, the target proteins were enriched at 35 kDa.

(4) FIG. 4 shows an electron microscope image of recombinant pET28a-PP-3K protein particles.

(5) FIG. 5 shows an electron microscope image of recombinant pET26b-PP-3C protein particles.

(6) FIGS. 6A to 6H show the levels of IgG antibodies against different antigens produced in mice immunized with phosphorylated polypeptide antigen vaccines in combination with Freund's adjuvant. FIG. 6A shows the levels of anti-Tau14-22 (pY18) IgG antibodies produced in WT mice immunized with the vaccines A1 to A16; each group contained 6 mice; each group of mice was immunized four times at two-week intervals using complete Freund's adjuvant for the first immunization and incomplete Freund's adjuvant for the second, third and fourth immunization; two weeks after the fourth immunization, the mice were subjected to blood sampling to obtain mouse serum for ELISA experiment; the results were expressed as the mean O.D.+SD values obtained by each group of mice. FIG. 6B shows the levels of anti-Tau198-209 (pS202/pT205) IgG antibodies produced in WT mice immunized with the vaccines A1 to A16; each group contained 6 mice; each group of mice was immunized four times at two-week intervals using complete Freund's adjuvant for the first immunization and incomplete Freund's adjuvant for the second, third and fourth immunization; two weeks after the fourth immunization, the mice were subjected to blood sampling to obtain mouse serum for ELISA experiment; the results were expressed as the mean O.D.+SD values obtained by each group of mice. FIG. 6C shows the levels of anti-Tau208-218 (pT212/pS214) IgG antibodies produced in WT mice immunized with the vaccines A1 to A16; each group contained 6 mice; each group of mice was immunized four times at two-week intervals using complete Freund's adjuvant for the first immunization and incomplete Freund's adjuvant for the second, third and fourth immunization; two weeks after the fourth immunization, the mice were subjected to blood sampling to obtain mouse serum for ELISA experiment; the results were expressed as the mean O.D.+SD values obtained by each group of mice. FIG. 6D shows the levels of anti-Tau227-239 (pS231/pS235) IgG antibodies produced in WT mice immunized with the vaccines A1 to A16; each group contained 6 mice; each group of mice was immunized four times at two-week intervals using complete Freund's adjuvant for the first immunization and incomplete Freund's adjuvant for the second, third and fourth immunization; two weeks after the fourth immunization, the mice were subjected to blood sampling to obtain mouse serum for ELISA experiment; the results were expressed as the mean O.D.+SD values obtained by each group of mice. FIG. 6E shows the levels of anti-Tau234-242 (pS238) IgG antibodies produced in WT mice immunized with the vaccines A1 to A16; each group contained 6 mice; each group of mice was immunized four times at two-week intervals using complete Freund's adjuvant for the first immunization and incomplete Freund's adjuvant for the second, third and fourth immunization; two weeks after the fourth immunization, the mice were subjected to blood sampling to obtain mouse serum for ELISA experiment; the results were expressed as the mean O.D.+SD values obtained by each group of mice. FIG. 6F shows the levels of anti-Tau258-266 (pS262) IgG antibodies produced in WT mice immunized with the vaccines A1 to A16; each group contained 6 mice; each group of mice was immunized four times at two-week intervals using complete Freund's adjuvant for the first immunization and incomplete Freund's adjuvant for the second, third and fourth immunization;

two weeks after the fourth immunization, the mice were subjected to blood sampling to obtain mouse serum for ELISA experiment; the results were expressed as the mean O.D.+SD values obtained by each group of mice. FIG. 6G shows the levels of anti-Tau392-400 (pS396) IgG antibodies produced in WT mice immunized with the vaccines A1 to A16; each group contained 6 mice; each group of mice was immunized four times at two-week intervals using complete Freund's adjuvant for the first immunization and incomplete Freund's adjuvant for the second, third and fourth immunization; two weeks after the fourth immunization, the mice were subjected to blood sampling to obtain mouse serum for ELISA experiment; the results were expressed as the mean O.D.+SD values obtained by each group of mice. FIG. 6H shows the levels of anti-Tau401-408 (pS404) IgG antibodies produced in WT mice immunized with the vaccines A1 to A16; each group contained 6 mice; each group of mice was immunized four times at two-week intervals using complete Freund's adjuvant for the first immunization and incomplete Freund's adjuvant for the second, third and fourth immunization; two weeks after the fourth immunization, the mice were subjected to blood sampling to obtain mouse serum for ELISA experiment; the results were expressed as the mean O.D.+SD values obtained by each group of mice.

(7) FIGS. 7A to 7S show the results of immunization of mice with phosphorylated polypeptide antigens coupled with a norovirus P protein through different immunization routes. Each group contained 6 mice. Each group of mice was immunized four times at two-week intervals. In the fourth week, sixth week and eighth week after the first immunization, the mice were subjected to blood sampling to obtain serum for ELISA experiment. FIG. 7A shows the results of intramuscular immunization of WT mice with A1 coupled with a norovirus P protein; the contents of anti-Tau14-22 (pY18) IgG antibodies and anti-Tau198-209 (pS202/pT205) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7B shows the results of intramuscular immunization of WT mice with A5 coupled with a norovirus P protein; the contents of anti-Tau14-22 (pY18) IgG antibodies, anti-Tau234-242 (pS238) IgG antibodies and anti-Tau258-266 (pS262) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7C shows the results of intramuscular immunization of WT mice with A9 coupled with a norovirus P protein; the contents of anti-Tau198-209 (pS202/pT205) IgG antibodies, anti-Tau234-242 (pS238) IgG antibodies and anti-Tau258-266 (pS262) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7D shows the results of intramuscular immunization of WT mice with A10 coupled with a norovirus P protein; the contents of anti-Tau198-209 (pS202/pT205) IgG antibodies, anti-Tau392-400 (pS396) IgG antibodies and anti-Tau401-408 (pS404) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7E shows the results of intramuscular immunization of WT mice with A11 coupled with a norovirus P protein; the contents of anti-Tau198-209 (pS202/pT205) IgG antibodies, anti-Tau392-400 (pS396) IgG antibodies and anti-Tau401-408 (pS404) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7F shows the results of intramuscular immunization of WT mice with A12 coupled with a norovirus P protein; the contents of anti-Tau208-218 (pT212/pS214) IgG antibodies and anti-Tau227-239 (pS231/pS235) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7G shows the results of intramuscular immunization of WT mice with A15 coupled with a norovirus P protein; the contents of anti-Tau234-242 (pS238) IgG antibodies, anti-Tau258-266 (pS262) IgG antibodies, anti-Tau392-400 (pS396) IgG antibodies and anti-Tau401-408 (pS404) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG.

7H shows the results of intraperitoneal immunization of WT mice with A2 coupled with a norovirus P protein; the contents of anti-Tau14-22 (pY18) IgG antibodies and anti-Tau198-209 (pS202/pT205) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7J shows the results of intraperitoneal immunization of WT mice with A3 coupled with a norovirus P protein; the contents of anti-Tau14-22 (pY18) IgG antibodies and anti-Tau208-218 (pT212/pS214) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7K shows the results of intraperitoneal immunization of WT mice with A7 coupled with a norovirus P protein; the contents of anti-Tau198-209 (pS202/pT205) IgG antibodies and anti-Tau227-239 (pS231/pS235) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7L shows the results of intraperitoneal immunization of WT mice with A11 coupled with a norovirus P protein; the contents of anti-Tau198-209 (pS202/pT205) IgG antibodies, anti-Tau392-400 (pS396) IgG antibodies and anti-Tau401-408 (pS404) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7M shows the results of intraperitoneal immunization of WT mice with A14 coupled with a norovirus P protein; the contents of anti-Tau208-218 (pT212/pS214) IgG antibodies, anti-Tau392-400 (pS396) IgG antibodies and anti-Tau401-408 (pS404) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7N shows the results of subcutaneous immunization of WT mice with A4 coupled with a norovirus P protein; the contents of anti-Tau14-22 (pY18) IgG antibodies and anti-Tau227-239 (pS231/pS235) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7O shows the results of subcutaneous immunization of WT mice with A6 coupled with a norovirus P protein; the contents of anti-Tau14-22 (pY18) IgG antibodies, anti-Tau392-400 (pS396) IgG antibodies and anti-Tau401-408 (pS404) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7P shows the results of subcutaneous immunization of WT mice with A8 coupled with a norovirus P protein; the contents of anti-Tau198-209 (pS202/pT205) IgG antibodies and anti-Tau227-239 (pS231/pS235) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7Q shows the results of subcutaneous immunization of WT mice with A11 coupled with a norovirus P protein; the contents of anti-Tau198-209 (pS202/pT205) IgG antibodies, anti-Tau392-400 (pS396) IgG antibodies and anti-Tau401-408 (pS404) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7R shows the results of subcutaneous immunization of WT mice with A13 coupled with a norovirus P protein; the contents of anti-Tau208-218 (pT212/pS214) IgG antibodies, anti-Tau392-400 (pS396) IgG antibodies and anti-Tau401-408 (pS404) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum. FIG. 7S shows the results of subcutaneous immunization of WT mice with A16 coupled with a norovirus P protein; the contents of anti-Tau234-242 (pS238) IgG antibodies, anti-Tau258-266 (pS262) IgG antibodies, anti-Tau392-400 (pS396) IgG antibodies and anti-Tau401-408 (pS404) IgG antibodies in the mouse serum were detected by ELISA method; the results were expressed as the mean O.D.+SD values obtained at a dilution of 1/200 of the mouse serum.

(8) FIG. 8 show that a phosphorylated polypeptide antigen coupled with PP-3C was combined respectively with CpG, AS02, AS03, MF59 and AS02+CpG adjuvant and a phosphorylated polypeptide antigen coupled with PP-3K was combined with AS02+CpG adjuvant to form different

combinations; wild mice were intramuscularly immunized respectively with the above different combinations at week 0, week 2, week 4, week 6 and week 12; blood was taken two weeks after every immunization to obtain serum. The phosphorylated polypeptide of SEQ ID NO: 1354 corresponding to phosphorylated amino acids at sites 202, 205, 396, and 404 of full-length Tau protein, was used as the coating antigen for ELISA experiment. The results were expressed as the concentrations (standardized by AT8 antibodies) of the antibodies of anti-phosphorylated polypeptide obtained in each group of mice.

(9) FIG. 9A to 9F show the analysis of immunogenicity of phosphorylated polypeptide antigen A5 coupled with PP-3K and combined with AS02+CpG adjuvant in mice. FIG. 9A shows the results of immunization of WT mice with phosphorylated polypeptide antigen A5 coupled with PP-3K and combined with AS02+CpG adjuvant; the content of antibodies in the mouse serum against the phosphorylated polypeptide with a sequence of SEQ ID NO: 1355 was detected. The results were expressed as the mean O.D.+SD value obtained at different dilutions of the mouse serum. FIG. 9B shows the results of immunization of P301S mice with phosphorylated polypeptide antigen A5 coupled with PP-3K and combined with AS02+CpG adjuvant; the content of antibodies in the mouse serum against the phosphorylated polypeptide with a sequence of SEQ ID NO: 1355 was detected. The results were expressed as the mean O.D.+SD value obtained at different dilutions of the mouse serum. FIG. 9C shows the results of immunization of P301S mice with phosphorylated polypeptide antigen A11 coupled with PP-3C and combined with AS02+CpG adjuvant; the content of antibodies in the mouse serum against the phosphorylated polypeptide with a sequence of SEQ ID NO: 1354 was detected. The results were expressed as the mean O.D.+SD value obtained at different dilutions of the mouse serum. FIG. 9D shows the results of immunization of P301S mice with phosphorylated polypeptide antigen A11 coupled with PP-3K and combined with AS02+CpG adjuvant; the content of antibodies in the mouse serum against the phosphorylated polypeptide with a sequence of SEQ ID NO: 1354 was detected. The results were expressed as the mean O.D.+SD value obtained at different dilutions of the mouse serum. FIG. 9E shows the results of immunization of WT mice with phosphorylated polypeptide antigen A12 coupled with PP-3K and combined with AS02+CpG adjuvant; the content of antibodies in the mouse serum against the phosphorylated polypeptide with a sequence of SEQ ID NO: 1356 was detected. The results were expressed as the mean O.D.+SD value obtained at different dilutions of the mouse serum. FIG. 9F shows the results of immunization of P301S mice with phosphorylated polypeptide antigen A12 coupled with PP-3K and combined with AS02+CpG adjuvant; the content of antibodies in the mouse serum against the phosphorylated polypeptide with a sequence of SEQ ID NO: 1356 was detected. The results were expressed as the mean O.D.+SD value obtained at different dilutions of the mouse serum.

(10) FIGS. 10A to 10H show the cellular immunity of the vaccines in P301S transgenic mice. FIG. 10A shows the ELISPOT results for PP-3K-A5 complex vaccine in P301S transgenic mice; the results were expressed as the number of IFN- γ spots generated per 10⁶ cells when the animal spleen cells in different immune groups were stimulated by a prokaryotically expressed full-length Tau protein stimulus. FIG. 10B shows the ELISPOT results for PP-3K-A5 complex vaccine in P301S transgenic mice; the results were expressed as the number of IFN- γ spots generated per 10⁶ cells when the animal spleen cells in different immune groups were stimulated by a nonphosphorylated A5 polypeptide stimulus. FIG. 10C shows the ELISPOT results for PP-3K-A11 complex vaccine in P301S transgenic mice; the results were expressed as the number of IFN- γ spots generated per 10⁶ cells when the animal spleen cells in different immune groups were stimulated by a prokaryotically expressed full-length Tau protein stimulus. FIG. 10D shows the ELISPOT results for PP-3K-A11 complex vaccine in P301S transgenic mice; the results were expressed as the number of IFN- γ spots generated per 10⁶ cells when the animal spleen cells in different immune groups were stimulated by a nonphosphorylated A11 polypeptide stimulus. FIG. 10E shows the ELISPOT results for PP-3C-A11 complex vaccine in P301S transgenic mice;

the results were expressed as the number of IFN- γ spots generated per 10^{sup.6} cells when the animal spleen cells in different immune groups were stimulated by a prokaryotically expressed full-length Tau protein stimulus. FIG. **10F** shows the ELISPOT results for PP-3C-A11 complex vaccine in P301S transgenic mice; the results were expressed as the number of IFN- γ spots generated per 10^{sup.6} cells when the animal spleen cells in different immune groups were stimulated by a nonphosphorylated A11 polypeptide stimulus. FIG. **10G** shows the ELISPOT results for PP-3K-A12 complex vaccine in P301S transgenic mice; the results were expressed as the number of IFN- γ spots generated per 10^{sup.6} cells when the animal spleen cells in different immune groups were stimulated by a prokaryotically expressed full-length Tau protein stimulus. FIG. **10H** shows the ELISPOT results for PP-3K-A12 complex vaccine in P301S transgenic mice; the results were expressed as the number of IFN- γ spots generated per 10^{sup.6} cells when the animal spleen cells in different immune groups were stimulated by a nonphosphorylated A12 polypeptide stimulus.

(11) FIGS. **11A** to **11D** show the rota-rod behavior of the vaccines in P301S transgenic mice. FIG. **11A** shows the results of the rota-rod experiment for PP-3K-A5 complex vaccine in P301S transgenic mice; the results are expressed as curves of the average time until mice fell from the rota-rod within 300 s versus the detection time points for animals in different immune groups. FIG. **11B** shows the results of the rota-rod experiment for PP-3C-A11 complex vaccine in P301S transgenic mice; the results are expressed as curves of the average time until mice fell from the rota-rod within 300 s versus the detection time points for animals in different immune groups. FIG. **11C** shows the results of the rota-rod experiment for PP-3K-A11 complex vaccine in P301S transgenic mice; the results are expressed as curves of the average time until mice fell from the rota-rod within 300 s versus the detection time points for animals in different immune groups. FIG. **11D** shows the results of the rota-rod experiment for PP-3K-A12 complex vaccine in P301S transgenic mice; the results are expressed as curves of the average time until mice fell from the rota-rod within 300 s versus the detection time points for animals in different immune groups.

(12) FIGS. **12A** to **12D** show the nesting behavior of the vaccines in P301S transgenic mice. FIG. **12A** shows the results of the nesting test for PP-3K-A5 complex vaccine in P301S transgenic mice; the results are expressed as the average of the scores given by three persons for each mouse in the nesting test. The results show that the mice in the immune group are statistically significantly different from the PBS group. FIG. **12B** shows the results of the nesting test for PP-3C-A11 complex vaccine in P301S transgenic mice; the results are expressed as the average of the scores given by three persons for each mouse in the nesting test. The results show that the mice in the immune group are statistically significantly different from the PBS group. FIG. **12C** shows the results of the nesting test for PP-3K-A11 complex vaccine in P301S transgenic mice; the results are expressed as the average of the scores given by three persons for each mouse in the nesting test. The results show that the mice in the immune group are statistically significantly different from the PBS group. FIG. **12D** shows the results of the nesting test for PP-3K-A12 complex vaccine in P301S transgenic mice; the results are expressed as the average of the scores given by three persons for each mouse in the nesting test. The results show that the mice in the immune group are statistically significantly different from the PBS group.

(13) FIGS. **13A** to **13D** show schematic diagrams of the results of HPLC quantification of the phosphorylated polypeptides coupled with PP-3C/C-PA-BLP. FIG. **13A** shows the HPLC diagram when the concentration of the standard was 0.03125 mg/mL; the peak labelled in the figure is the peak position of the standard. FIG. **13B** exemplarily shows the HPLC diagram of the phosphorylated polypeptide dissociated from the phosphorylated polypeptide sample coupled with PP-3C prepared in Example 4-2; the peak labelled in the figure is the peak position of the sample. FIG. **13C** exemplarily shows the HPLC diagram of the phosphorylated polypeptide dissociated from the phosphorylated polypeptide sample coupled with C-PA-BLP prepared in Example 33; the peak labelled in the figure is the peak position of the sample. FIG. **13D** shows a graph of a standard curve of the concentration of the standard versus the peak area.

(14) FIG. 14 show is a schematic diagram of pET28a-C-PA plasmid that has been constructed.

(15) FIGS. 15A to 15B show the recombinant pET28a-C-PA plasmid construct and the results of purification and expression. FIG. 15A is a schematic diagram of identification of pET28a-C-PA plasmid that has been constructed by double-enzyme digestion; as shown in FIG. 15A, the target band is at 700 bp. FIG. 15B is a SDS-PAGE graph of a C-PA protein sample; as shown in FIG. 15B, the protein band appeared around 24 kDa.

DETAILED DESCRIPTION OF THE INVENTION

(16) The content of the present invention is illustrated below through specific embodiments. It should be understood that the specific embodiments are for illustrative purposes only, and do not mean that the content of the present invention is limited to the specific embodiments.

Example 1: Preparation of Phosphorylated Polypeptide Antigen Vaccines

(17) In this example, based on human full-length Tau protein (SEQ ID NO: 1332), the inventor designed the following 15 phosphorylated polypeptide antigen vaccines comprising two polypeptide fragments from the human full-length Tau protein; the specific sequence information of the 15 phosphorylated polypeptide antigen vaccines is shown in Table 1 below:

(18) TABLE-US-00001 TABLE 1 Sequence information of the 15 phosphorylated polypeptide antigen vaccines

The corresponding full-length Sequence	Vaccine Tau protein fragments	No. No. Sequences phosphorylation sites	SEQ ID
HAGTYGLGDSSPGSP	Tau14-22	[pY18], Tau198- NO: 201	GTPGSRC 209[pS202, pT205] SEQ ID A1
HAGTYGLGDRSGYSS	Tau14-22	[pY18], Tau194- NO: 225	PGSPGTPGSRSRTPC 213[pS202, pT205] SEQ ID A2
HAGTYGLGDSRSRTP	Tau14-22	[pY18], Tau208- NO: 306	SLPTPC 218[pT212, pS214] SEQ ID A3
HAGTYGLGDAVVRTPP	Tau14-22	[pY18], Tau227- NO: 387	KSPSSAC 239[pT231, pS235] SEQ ID A4
HAGTYGLGDKSPSSA	Tau14-22	[pY18], Tau234- NO: 468	KRSRKIGSTENLC 242[pS238], Tau258- 266[pS262] SEQ ID A5
HAGTYGLGDIVYKSPV	Tau14-22	[pY18], Tau392- NO: 558	VSGDTPRHLC 408[pS396, pS404] SEQ ID A6
SSPGSPGTPGSRAVV	Tau198-209	[pS202, pT205], NO: 567	RTPPKSPSSAC Tau227- 239[pT231, pS235] SEQ ID A7
SGYSSPGSPGTPGSR	Tau195-213	[pS202, pT205], NO: 769	SRTPVVTPPKSPSSAC Tau228-239[pT231, pS235] SEQ ID A8
SSPGSPGTPGSRKSP	Tau198-209	[pS202, pT205], NO: 784	SSAKRSRKIGSTENLC Tau234-242[pS238], Tau258- 266[pS262] SEQ ID A9
SPVVSGDTPRHLC	Tau395-406	[pS396, pS404] SEQ ID A10	SSPGSPGTPGSRIVYK Tau198-209[pS202, pT205], NO: 875
SGYSSPGSPGTPGSR	Tau195-213	[pS202, pT205], NO: 1020	SRTPKSPVVSGDTPRC Tau395-406[pS396, pS404] SEQ ID A11
SRSRTPSLPTPAVVRT	Tau208-218	[pT212, pS214], NO: 1101	PPKSPSSAC Tau227-239[pT231, pS235] SEQ ID A12
SRSRTPSLTPKSPSS	Tau208-218	[pT212, pS214], NO: 1182	AKRSRKIGSTENLC Tau234-242[pS238], Tau258- 266[pS262] SEQ ID A13
SRSRTPSLTPIVYKSP	Tau208-218	[pT212, pS214], NO: 1272	VVSGDTPRHLC Tau392-408[pS396, pS404] SEQ ID A14
SPSSAKRSRKIGSTEN	Tau235-242	[pS238], Tau258- NO: 1313	VYKSPVVSGDTPRH 265[pS262], Tau393- C 407[pS396, pS404] SEQ ID A15
KSPSSAKRSRKIGSTE	Tau234-242	[pS238], Tau258- NO: 1330	NLKSPVVSGDTPRC 266[pS262], Tau395- 406[pS396, pS404] SEQ ID A16

(19) The aforementioned 15 phosphorylated polypeptide antigen vaccines were synthesized and prepared by GL biochem (Shanghai) Ltd., and are presented in lyophilized form.

Example 2: Preparation of a Cysteine-Modified Norovirus P Protein (PP-3C)

(20) The preparation of norovirus P protein plasmid using pET26b(+) as a vector has been described in Chinese invention patent application 2015104155561. In this example, three point mutations were performed using this plasmid as a template, one amino acid in each of the three loop structures of the norovirus P particle was mutated to a cysteine, and the protein was expressed and purified.

(21) 2.1 Construction of PP-3C Plasmid

(22) Three rounds of site-directed mutagenesis were performed by site-directed mutation method using the pET26b-P protein plasmid which has already been constructed in Chinese invention patent application 2015104155561. Respectively, 5'ATCGCTGGAACACAA3' in loop1 was mutated to 5'ATCGCTTGCACACAA3'. The specific method was as follows:

(23) TABLE-US-00002 SEQ ID NO: 1334 (forward):

CTGTGAACATCGCTACTTTCCGCGGCGACGTCACACACATCGCTTGCAC

ACAAACTAC, and SEQ ID NO: 1335 (reverse):

GTAGTTTTGTGTGCAAGCGATGTGTGTGACGTCGCCGCGGAAAGTAGCG

ATGTTACAG;

the entire plasmid was subjected to PCR reaction, wherein the PCR reaction system was a KOD-Plus DNA polymerase system (purchased from TOYOBO Corporation), and the total volume of the reaction system was 50 μ L (5 μ L buffer, 0.2 mM dNTP, 1 mM magnesium sulfate, 0.3 μ M upstream primer and 0.3 μ M downstream primer, 50 ng template DNA, 1 μ L KOD enzyme, and water which was added to a final volume of 50 μ L). PCR was carried out in accordance with the reaction system instructions to obtain 20 μ L of PCR products. 1 μ L of DpnI enzyme (purchased from NEB Corporation) was added to the PCR products. Digestion was carried out at 37° C. for 1 h. 10 μ L of the digested products were added to Tran1-Blue competent cells (purchased from Beijing TransGen Biotech, Ltd.), and placed on ice for 30 min. After that, the cells were subjected to heat shock at 42° C. for 45 s, placed on ice for 2 min and then were added to 600 μ L of fluid LB medium without resistance, and recovered at 200 rpm at 37° C. for 1 h. The culture broth was plated on a LB solid culture plate containing kanamycin (15 μ g/mL), and were placed upside down at 37° C. overnight. The mutant plasmid clones were obtained, and the sequence was verified by sequencing.

(24) The plasmid obtained by the first round of point mutation was subjected to a second round of point mutation. 5'ACCTCAAACGAT3' in loop2 was mutated to 5'ACCTGCAACGA3'. The specific method was as follows:

(25) TABLE-US-00003 SEQ ID NO: 1336 (forward):

GCAATTCAGCACAGACACCTGCAACGATTTCGAGACTGGCC, and SEQ ID NO:

1337 (reverse): GGCCAGTCTCGAAATCGTTGCAGGTGTCTGTGCTGAATTGC;

the entire plasmid was subjected to PCR reaction, wherein the PCR reaction system was a KOD-Plus DNA polymerase system (purchased from TOYOBO Corporation), and the total volume of the reaction system was 50 μ L (5 μ L buffer, 0.2 mM dNTP, 1 mM magnesium sulfate, 0.3 μ M upstream primer and 0.3 μ M downstream primer, 50 ng template DNA, 1 μ L KOD enzyme, and water which was added to a final volume of 50 μ L). PCR was carried out in accordance with the reaction system instructions to obtain 20 μ L of PCR products. 1 μ L of DpnI enzyme (purchased from NEB Corporation) was added to the PCR products. Digestion was carried out at 37° C. for 1 h. 10 μ L of the digested products were added to Tran1-Blue competent cells (purchased from Beijing TransGen Biotech, Ltd.), and placed on ice for 30 min. After that, the cells were subjected to heat shock at 42° C. for 45 s, placed on ice for 2 min. and then were added to 600 μ L of fluid LB medium without resistance, and recovered at 200 rpm at 37° C. for 1 h. The culture broth was plated on a LB solid culture plate containing kanamycin (15 μ g/mL), and were placed upside down at 37° C. overnight. The mutant plasmid clones were obtained, and the sequence was verified by sequencing.

(26) The plasmid obtained by the second round of point mutation was subjected to a third round of point mutation. 5'GACGGCAGCACCC3' in loop3 was mutated to 5'GACTGCAGCACCC3'. The specific method was as follows:

(27) TABLE-US-00004 SEQ ID NO: 1338 (forward):

CCGTGGGTGTGTTCAAGACTGCAGCACCCTCACCAGAACG, and SEQ ID NO:

1339 (reverse): CGTTCTGGTGAGTGGTGCTGCAGTCTTGAACGACACCCACGG;

(28) the entire plasmid was subjected to PCR reaction, wherein the PCR reaction system was a

KOD-Plus DNA polymerase system (purchased from TOYOBO Corporation), and the total volume of the reaction system was 50 μ L (5 μ L buffer, 0.2 mM dNTP, 1 mM magnesium sulfate, 0.3 μ M upstream primer and 0.3 μ M downstream primer, 50 ng template DNA, 1 μ L KOD enzyme, and water which was added to a final volume of 50 μ L). PCR was carried out in accordance with the reaction system instructions to obtain 20 μ L of PCR products. 1 μ L of DpnI enzyme (purchased from NEB Corporation) was added to the PCR products. Digestion was carried out at 37° C. for 1 h. 10 μ L of the digested products were added to Tran1-Blue competent cells (purchased from Beijing TransGen Biotech, Ltd.), and placed on ice for 30 min. After that, the cells were subjected to heat shock at 42° C. for 45 s, placed on ice for 2 min. and then were added to 600 μ L of fluid LB medium without resistance, and recovered at 200 rpm at 37° C. for 1 h. The culture broth was plated on a LB solid culture plate containing kanamycin (15 μ g/mL), and were placed upside down at 37° C. overnight. The mutant plasmid clones were obtained, and the sequence was verified by sequencing. The resulting plasmid of pET26b-PP-3C is shown in FIG. 1B.

(29) 2.2 Construction of the P Protein that has been Successfully Mutated, Namely the PP-3C Gene, on a pCold IV Carrier

(30) An EcoR I restriction site was introduced at the 5' end of the PP-3C gene (whose sequence is represented by SEQ ID NO: 1358) by PCR method, and a Hind III restriction site was introduced at the 3' end. The specific method was as follows:

(31) TABLE-US-00005 SEQ ID NO: 1340 (forward):

GGGAATTCCATATGAAGCCCTTCTCGGTCCCTATCCTGACAG, and SEQ ID NO: 1341 (reverse): CCCAAGCTTTTACAAGGCTCTGCGACGACCGGCTC;

the pET26b-PP-3C plasmid obtained above was subjected to PCR reaction, wherein the PCR reaction system was a KOD-Plus DNA polymerase system (purchased from TOYOBO Corporation), and the total volume of the reaction system was 50 μ L (5 μ L buffer, 0.2 mM dNTP, 1 mM magnesium sulfate, 0.3 μ M upstream primer and 0.3 μ M downstream primer, 50 ng template DNA, 1 μ L KOD enzyme, and water which was added to a final volume of 50 μ L). PCR was carried out in accordance with the reaction system instructions to obtain 50 μ L of PCR products. The products were subjected to agarose gel electrophoresis, and the fragments of interest were recovered using gel recovery kit (purchased from Tiangen Biotech Co., Ltd.) to obtain a PP-3C gene fragment with an EcoRI restriction site at the 5' end and a Hind III restriction site at the 3' end.

(32) 1 μ L of Hind III enzyme and 1 μ L of EcoR I enzyme (purchased from Takara Corporation) and 5 μ L of enzyme digestion buffer (purchased from Takara Corporation) were added respectively to 2 μ g of the aforementioned recovered fragment, and finally sterile water was added to the system to reach a final volume of 50 μ L. Digestion was carried out at 37° C. for 2 hours. The products were purified using common DNA product purification kit (purchased from Tiangen Biotech Co., Ltd.) to obtain the fragment of interest having sticky ends.

(33) 1 μ L of Hind III enzyme and 1 μ L of EcoR I enzyme (purchased from Takara Corporation) and 5 μ L of enzyme digestion buffer (purchased from Takara Corporation) were added respectively to 2 μ g of pCold IV vector plasmid (purchased from Novagen Corporation), and finally sterile water was added to the system to reach a final volume of 50 μ L. Digestion was carried out at 37° C. for 2 hours. The products were subjected to agarose gel electrophoresis, and the vector fragments were recovered using gel recovery kit (purchased from Tiangen Biotech Co., Ltd.) to obtain a plasmid vector having double enzyme-digested sticky ends

(34) The above double enzyme-digested vector fragment and fragment of interest (with a molar ratio of 1:3, and a total volume of 15 μ L) were mixed, and 0.75 μ L of T4 ligase (purchased from Takara Corporation) and 1.5 μ L of ligase buffer (purchased from Takara Corporation) were added. The ligation was carried out at 16° C. overnight. 10 μ L of the ligated products were added to Tran1-Blue competent cells (purchased from Beijing TransGen Biotech, Ltd.), and placed on ice for 30 min. After that, the cells were subjected to heat shock at 42° C. for 45 s, placed on ice for 2

min, and then were added to 600 μ L of fluid LB medium without resistance, and recovered at 200 rpm at 37° C. for 1 h. The culture broth was plated on a LB solid culture plate containing ampicillin (15 μ g/mL), and were placed upside down at 37° C. overnight. The recombinant clones were obtained, and the sequence was verified by sequencing. A pColdIV-PP-3C plasmid that can express PP-3C protein was obtained, as shown in FIG. 1E.

(35) 2.3 Expression of the Cysteine-Modified Norovirus P Protein

(36) 1 μ L of the recombinant plasmids prepared in the above examples were respectively added to 100 μ L of *E. coli* BL21 competent cells (purchased from TransGen Corporation), ice-bathed for 30 min, heat shocked for 90 s in a water-bath at 42° C., and then ice-bathed for 2 min. 600 μ L of LB medium was added to the mixture, and cultured at 180 rpm/min at 37° C. for 1 h. The mixture was coated evenly on a LB solid medium containing kanamycin (15 μ g/mL) resistance and cultured at 37° C. for 24 h to obtain strains that can stably express recombinant proteins. A growing colony was picked and inoculated into 20 mL of LB medium. The mixture was cultured at 220 rpm at 37° C. When the OD value of the culture mixture reached 1.0, induction by Isopropyl β -D-Thiogalactoside (IPTG at a final concentration of 0.5 mmol/L) was carried out at 220 rpm at 16° C. overnight. After the induction, the culture broth was centrifuged at 4,000 rpm for 20 min. The supernatant was discarded, and the bacterial precipitates were resuspended with PBS.

Centrifugation was conducted again at 4,000 rpm for 20 min and the supernatant was discarded to obtain the bacterial precipitates containing proteins of interest.

(37) 2.4 Purification of the Cysteine-Modified Norovirus P Protein

(38) The bacterial precipitates obtained in 2.3 were resuspended by adding 20 mL of protein buffer (pH8.0, containing 50 mM Tris and 300 mM KCl). The bacteria were lysed by sonication on ice for 30 min. The mixture was centrifuged at 12,000 rpm at 4° C. for 30 min. Subsequently, the supernatant was taken and allowed to pass through 0.45 μ m filter membrane to obtain crude extracts of proteins.

(39) The structure of the recombinant PP-3C particle protein is shown in FIG. 2A.

(40) The crude extracts of proteins were purified using an anion exchange column (purchased from GE Corporation). The specific scheme was as follows: First, the exchange column was rinsed with ultrapure water in a volume of about 100 mL, followed by equilibration with PB solution (pH 5.0) at a flow rate of 2 mL/min. Then 20 mL of the crude extracts of proteins were added to the exchange column at a flow rate of 1 mL/min. After the sample was completely loaded onto the column, the exchange column was rinsed with PB solution (pH 7.0) to remove hybrid protein, and then eluted with PB solution containing 0.5 mol/L NaCl. The proteins at peak value were collected to obtain the proteins of interest.

(41) The proteins were further purified by a hydrophobic chromatography column (purchased from GE Corporation). The specific scheme was as follows: First, the column was rinsed with ultrapure water, and then the hydrophobic column was rinsed with PB solution (pH 7.0) at a flow rate of 2 mL/min. After the column was equilibrated well, the protein sample was loaded onto it. After the sample was completely loaded onto the column, the column was eluted by gradient with PB (pH 7.0) and 1 mol/L NaCl solution for 2 hours. The concentration of NaCl decreases from 1 mol/L to 0.1 mol/L. The proteins were collected at peak value.

(42) The sizes of the resulting P particle protein monomers were identified by reductive SDS-PAGE.

(43) Then the tetracosamers of P protein particles were further isolated and purified using a Superdex 200 molecular sieve (purchased from GE Corporation). The procedures were as follows: The column was rinsed with ultrapure water at a flow rate of 1 mL/min for one volume of the column, and then again with about 120 mL of PB buffer (pH=5). After that, 2 mL of protein extraction solution was added to the column and washed with PB buffer at a flow rate of 1 mL/min. The proteins at peak value were collected to obtain the tetracosamers of P particle proteins. Three proteins were tested for their polymer structures by native polyacrylamide gel electrophoresis. As

NO: 1341 (reverse): GTTCATGGTGTAGTTTGTGTCTTTCCAGCGATGTGTGTGACGTC;
the entire plasmid was subjected to PCR reaction, wherein the PCR reaction system was a KOD-Plus DNA polymerase system (purchased from TOYOBO Corporation), and the total volume of the reaction system was 50 μ L (5 μ L buffer, 0.2 mM dNTP, 1 mM magnesium sulfate, 0.3 μ M upstream primer and 0.3 μ M downstream primer, 50 ng template DNA, 1 μ L KOD enzyme, and water which was added to a final volume of 50 μ L). PCR was carried out in accordance with the reaction system instructions to obtain 20 μ L of PCR products. 1 μ L of DpnI enzyme (purchased from NEB Corporation) was added to the PCR products. Digestion was carried out at 37° C. for 1 h. 10 μ L of the digested products were added to Tran1-Blue competent cells (purchased from Beijing TransGen Biotech, Ltd.), and placed on ice for 30 min. After that, the cells were subjected to heat shock at 42° C. for 45 s, placed on ice for 2 min, and then were added to 600 μ L of fluid LB medium without resistance, and recovered at 200 rpm at 37° C. for 1 h. The culture broth was plated on a LB solid culture plate containing kanamycin (15 μ g/mL), and were placed upside down at 37° C. overnight. The mutant plasmid clones were obtained, and the sequence was verified by sequencing.

(53) The PCR product obtained by the first round of site-directed mutation was subjected to a second round of site-directed mutation. 5' ACCTCAAACGAT 3' in loop2 was mutated to 5' ACCTCAAAGAACGAT3'. The specific method was as follows:

(54) TABLE-US-00008 SEQ ID NO: 1342 (forward):

CAATTCAGCACAGACACCTCAAAGAACGATTTTCGAGACTGGCCAG, and SEQ ID NO: 1343 (reverse):

CTGGCCAGTCTCGAAATCGTTCTTTGAGGTGTCTGTGCTGAATTG;

the entire plasmid was subjected to PCR reaction, wherein the PCR reaction system was a KOD-Plus DNA polymerase system (purchased from TOYOBO Corporation), and the total volume of the reaction system was 50 μ L (5 μ L buffer, 0.2 mM dNTP, 1 mM magnesium sulfate, 0.3 μ M upstream primer and 0.3 μ M downstream primer, 50 ng template DNA, 1 μ L KOD enzyme, and water which was added to a final volume of 50 μ L). PCR was carried out in accordance with the reaction system instructions to obtain 20 μ L of PCR products. 1 μ L of DpnI enzyme (purchased from NEB Corporation) was added to the PCR products. Digestion was carried out at 37° C. for 1 h. 10 μ L of the digested products were added to Tran1-Blue competent cells (purchased from Beijing TransGen Biotech, Ltd.), and placed on ice for 30 min. After that, the cells were subjected to heat shock at 42° C. for 45 s, placed on ice for 2 min, and then were added to 600 μ L of fluid LB medium without resistance, and recovered at 200 rpm at 37° C. for 1 h. The culture broth was plated on a LB solid culture plate containing kanamycin (15 μ g/mL), and were placed upside down at 37° C. overnight. The mutant plasmid clones were obtained, and the sequence was verified by sequencing.

(55) The PCR product obtained by the second round of site-directed mutation was subjected to a third round of site-directed mutation. 5' GACGGCAGCACC 3' in loop3 was mutated to 5' GACGGCAGCACC3'. The specific method was as follows:

(56) TABLE-US-00009 SEQ ID NO: 1344 (forward):

GTGGGTGTCGTTCAAGACGGCAAGAGCACCCTCACCAGAACGAA, and SEQ ID NO: 1345 (reverse):

TTCGTTCTGGTGAGTGGTGCTCTTGCCGTCTTGAACGACACCCAC;

the entire plasmid was subjected to PCR reaction, wherein the PCR reaction system was a KOD-Plus DNA polymerase system (purchased from TOYOBO Corporation), and the total volume of the reaction system was 50 μ L (5 μ L buffer, 0.2 mM dNTP, 1 mM magnesium sulfate, 0.3 μ M upstream primer and 0.3 μ M downstream primer, 50 ng template DNA, 1 μ L KOD enzyme, and water which was added to a final volume of 50 μ L). PCR was carried out in accordance with the reaction system instructions to obtain 20 μ L of PCR products. 1 μ L of DpnI enzyme (purchased from NEB Corporation) was added to the PCR products. Digestion was carried out at 37° C. for 1

h. 10 μ L of the digested products were added to Tran1-Blue competent cells (purchased from Beijing TransGen Biotech, Ltd.), and placed on ice for 30 min. After that, the cells were subjected to heat shock at 42° C. for 45 s, placed on ice for 2 min. and then were added to 600 μ L of fluid LB medium without resistance, and recovered at 200 rpm at 37° C. for 1 h. The culture broth was plated on a LB solid culture plate containing kanamycin (15 μ g/mL), and were placed upside down at 37° C. overnight. The mutant plasmid clones were obtained, and the sequence was verified by sequencing (the sequence of the PP-3K gene is represented by SEQ ID NO: 1360). The pET28a-P protein-3K plasmid is shown as FIG. 1D.

3.3 Expression of the Lysine-Modified Norovirus P Protein

(57) 1 μ L of the recombinant plasmids prepared in the above examples were respectively added to 100 μ L of *E. coli* BL21 competent cells (purchased from TransGen Corporation), ice-bathed for 30 min, heat shocked for 90 s in a water-bath at 42° C., and then ice-bathed for 2 min. 600 μ L of LB medium was added to the mixture, and cultured at 180 rpm/min at 37° C. for 1 h. The mixture was coated evenly on a LB solid medium containing kanamycin (15 μ g/mL) resistance and cultured at 37° C. for 24 h to obtain strains that can stably express recombinant proteins. A growing colony was picked and inoculated into 20 mL of LB medium. The mixture was cultured at 220 rpm at 37° C. When the OD value of the culture mixture reached 1.0, induction by Isopropyl β -D-Thiogalactoside (IPTG at a final concentration of 0.5 mmol/L) was carried out at 220 rpm at 16° C. overnight. After the induction, the culture broth was centrifuged at 4,000 rpm for 20 min. The supernatant was discarded, and the bacterial precipitates were resuspended with PBS. Centrifugation was conducted again at 4,000 rpm for 20 min and the supernatant was discarded to obtain the bacterial precipitates containing proteins of interest.

(58) 3.4 Purification of the Lysine-Modified Norovirus P Protein

(59) The bacterial precipitates obtained in 3.3 were resuspended by adding 20 ml of protein buffer (pH8.0, containing 50 mM Tris and 300 mM KCl). The bacteria were lysed by sonication on ice for 30 min. The mixture was centrifuged at 12,000 rpm at 4° C. for 30 min. Subsequently, the supernatant was taken and allowed to pass through 0.45 μ m filter membrane to obtain crude extracts of proteins.

(60) The structure of the recombinant PP-3K particle is shown as FIG. 3A.

(61) The crude extracts of proteins were purified using an anion exchange column (purchased from GE Corporation). The specific scheme was as follows: First, the exchange column was rinsed with ultrapure water in a volume of about 100 mL, followed by equilibration with PB solution (pH 5.0) at a flow rate of 2 mL/min. Then 20 mL of the crude extracts of proteins were added to the exchange column at a flow rate of 1 mL/min. After the sample was completely loaded onto the column, the exchange column was rinsed with PB solution (pH 7.0) to remove hybrid protein, and then eluted with PB solution containing 0.5 mol/L NaCl. The proteins at peak value were collected to obtain the proteins of interest.

(62) The proteins were further purified by a hydrophobic chromatography column (purchased from GE Corporation). The specific scheme was as follows: First, the column was rinsed with ultrapure water, and then the hydrophobic column was rinsed with PB solution (pH 7.0) at a flow rate of 2 mL/min. After the column was equilibrated well, the protein sample was loaded onto it. After the sample was completely loaded onto the column, the column was eluted by gradient with PB (pH 7.0) and 1 mol/L NaCl solution for 2 hours. The concentration of NaCl decreases from 1 mol/L to 0.1 mol/L. The proteins were collected at peak value.

(63) The sizes of P particle protein monomers were identified by reductive SDS-PAGE. Then the tetracosamers of P protein particles were further isolated and purified using a Superdex 200 molecular sieve (purchased from GE Corporation). The procedures were as follows: The column was rinsed with ultrapure water at a flow rate of 1 mL/min for one volume of the column, and then again with about 120 mL of PB buffer (pH=5). After that, 2 mL of protein extraction solution was added to the column and washed with PB buffer at a flow rate of 1 mL/min. The proteins at peak

value were collected to obtain the tetracosamers of P particle proteins. Three proteins were tested for their polymer structures by native polyacrylamide gel electrophoresis, as shown in FIG. 2C. Protein bands are above 250 KDa, which indicates that recombinant P proteins can self-assemble into dodecamers and tetracosamers of P protein particles in vitro and maintain equilibrium; and the proteins can remain their polymer form after being purified. Electron microscopy analysis of purified PP-3K protein particles showed that PP-3K mostly formed large particle aggregates (as shown in FIG. 5).

Example 4: Coupling Reaction of PP-3C with a Phosphorylated Polypeptide Antigen Vaccine

(64) Example 4-1: The purified PP-3C was subjected to a buffer-exchange treatment with a desalting column from GE Corporation to change the buffer system of the protein to a 0.1 M ammonium bicarbonate solution (pH=7.5). The PP-3C after the buffer-exchange treatment was quantified, and was diluted to a concentration of 0.3 mg/mL. In the ratio of the molar amount of protein to the molar amount of polypeptide of 1:30, each of the phosphorylated polypeptide antigen vaccines in lyophilized form prepared in Example 1 was added and mixed slowly. The mixed system was incubated at 2° C. to 8° C. with slow shaking for 48 hours. The resulting product was concentrated by ultrafiltration to remove unbound polypeptides, and the buffer system was changed to a PBS buffer system.

(65) Example 4-2: The purified PP-3C was subjected to a buffer-exchange treatment with a desalting column from GE Corporation to change the buffer system of the protein to a 1 M ammonium bicarbonate solution (pH=8.0). The PP-3C after the buffer-exchange treatment was quantified, and was diluted to a concentration of 0.5 mg/mL. In the ratio of the molar amount of protein to the molar amount of polypeptide of 1:50, each of the phosphorylated polypeptide antigen vaccines in lyophilized form prepared in Example 1 was added and mixed slowly. The mixed system was incubated at 2° C. to 8° C. with slow shaking for 48 hours. The resulting product was concentrated by ultrafiltration to remove unbound polypeptides, and the buffer system was changed to a PBS buffer system.

(66) Example 4-3: The purified PP-3C was subjected to a buffer-exchange treatment with a desalting column from GE Corporation to change the buffer system of the protein to a 0.1 M ammonium bicarbonate solution (pH=7.8). The PP-3C after the buffer-exchange treatment was quantified, and was diluted to a concentration of 0.3 mg/mL. In the ratio of the molar amount of protein to the molar amount of polypeptide of 1:100, each of the phosphorylated polypeptide antigen vaccines in lyophilized form prepared in Example 1 was added and mixed slowly. The mixed system was incubated at 2° C. to 8° C. with slow shaking for 48 hours. The resulting product was concentrated by ultrafiltration to remove unbound polypeptides, and the buffer system was changed to a PBS buffer system.

(67) Example 4-4: The purified PP-3C was subjected to a buffer-exchange treatment with a desalting column from GE Corporation to change the buffer system of the protein to a 0.1 M ammonium bicarbonate solution (pH=8.8). The PP-3C after the buffer-exchange treatment was quantified, and was diluted to a concentration of 0.3 mg/mL. In the ratio of the molar amount of protein to the molar amount of polypeptide of 1:30, each of the phosphorylated polypeptide antigen vaccines in lyophilized form prepared in Example 1 was added and mixed slowly. The mixed system was incubated at 2° C. to 8° C. with slow shaking for 72 hours. The resulting product was concentrated by ultrafiltration to remove unbound polypeptides, and the buffer system was changed to a PBS buffer system.

Example 5: Coupling Reaction of PP-3K with a Phosphorylated Polypeptide Antigen Vaccine

(68) Example 5-1: The purified PP-3K solution was quantified by BCA and the pH was adjusted to 7.2. This protein solution was mixed with a 1 mg/mL sulfo-maleimide (sulfo-SMCC) solution in the ratio of the molar amount of SMCC to the molar amount of PP-3K of 5:1, and the mixture was reacted in a water bath at 25° C. for 30 min. The mixture was subjected to a desalting treatment with a desalting column from GE Corporation, the system was changed to a PBS solution (pH=7.2-

7.4), and unbound sulfo-SMCC was removed. The reaction product was taken and each of the lyophilized products of the phosphorylated polypeptide antigen vaccines prepared in Example 1 was added to the system in the ratio of the molar amount of PP-3K to the molar amount of phosphorylated polypeptide of 1:5, and mixed slowly. Then, the reaction system was reacted in a water bath at 25° C. for 30 min. The reaction product was collected, and the mixture was subjected to a desalting treatment with a desalting column from GE Corporation, the system was changed to a PBS solution (pH=7.2-7.4), and unbound phosphorylated polypeptides were removed.

(69) Example 5-2: The purified PP-3K solution was quantified by BCA. This protein solution was mixed with a 1 mg/mL sulfo-maleimide (sulfo-SMCC) solution in the ratio of the molar amount of SMCC to the molar amount of PP-3K of 20:1, and the mixture was reacted in a water bath at 25° C. for 30 min. The mixture was subjected to a desalting treatment with a desalting column from GE Corporation, the system was changed to a PB solution (pH=8.5), and unbound sulfo-SMCC was removed. The reaction product was taken and each of the lyophilized products of the phosphorylated polypeptide antigen vaccines prepared in Example 1 was added to the system in the ratio of the molar amount of PP-3K to the molar amount of phosphorylated polypeptides of 1:30, and mixed slowly. Then, the reaction system was reacted in a water bath at 25° C. for 30 min. The reaction product was collected, and the mixture was subjected to a desalting treatment with a desalting column from GE Corporation, the system was changed to a PB solution (pH=8.5), and unbound phosphorylated polypeptides were removed.

(70) Example 5-3: The purified PP-3K solution was quantified by BCA. This protein solution was mixed with a 1 mg/mL sulfo-maleimide (sulfo-SMCC) solution in the ratio of the molar amount of SMCC to the molar amount of PP-3K of 20:1, and the mixture was reacted in a water bath at 25° C. for 30 min. The mixture was subjected to a desalting treatment with a dialysis bag having a molecular weight cut-off of 20 kd (the dialysis volume ratio is 1,000:1), and was dialyzed overnight at 2° C. to 8° C., the system was changed to a PB solution (pH=7.0), and unbound sulfo-SMCC was removed. The reaction product was taken and each of the lyophilized products of the phosphorylated polypeptide antigen vaccines prepared in Example 1 was added to the system in the ratio of the molar amount of PP-3K to the molar amount of phosphorylated polypeptides of 1:50, and mixed slowly. Then, the reaction system was reacted in a water bath at 25° C. for 30 min. The mixture was subjected to a desalting treatment with a dialysis bag having a molecular weight cut-off of 20 kDa (the dialysis volume ratio is 1,000:1), and was dialyzed overnight at 2° C. to 8° C., the system was changed to a PB solution (pH=7.0), and unbound phosphorylated polypeptides were removed.

(71) Example 5-4: The purified PP-3K solution was quantified by BCA. This protein solution was mixed with a 1 mg/mL sulfo-maleimide (sulfo-SMCC) solution in the ratio of the molar amount of SMCC to the molar amount of PP-3K of 20:1, and the mixture was reacted with slow shaking overnight at 2° C. to 8° C. The mixture was subjected to a desalting treatment with a dialysis bag having a molecular weight cut-off of 20 kd (the dialysis volume ratio is 1,000:1), and was dialyzed overnight at 2° C. to 8° C., the system was changed to a PBS solution (pH=7.2-7.4), and unbound sulfo-SMCC was removed. The reaction product was taken and each of the lyophilized products of the phosphorylated polypeptide antigen vaccines prepared in Example 1 was added to the system in the ratio of the molar amount of PP-3K to the molar amount of phosphorylated polypeptides of 1:50, and mixed slowly. Then, the reaction system was reacted with slow shaking overnight at 2° C. to 8° C. The mixture was subjected to a desalting treatment with a dialysis bag having a molecular weight cut-off of 20 kd (the dialysis volume ratio is 1,000:1), and was dialyzed overnight at 2° C. to 8° C., the system was changed to a PBS solution (pH=7.2-7.4), and unbound phosphorylated polypeptides were removed.

(72) Example 5-5: The purified PP-3K solution was quantified by BCA. This protein solution was mixed with a 20 mg/ml dimethylformamide (DMF) solution of maleimide (SMCC) in the ratio of the molar amount of SMCC to the molar amount of PP-3K of 10:1, and the mixture was reacted in

a water bath at 25° C. for 30 min. The mixture was subjected to a desalting treatment with a desalting column from GE Corporation, the system was changed to a PBS solution (pH=7.2-7.4), and unbound SMCC was removed. The reaction product was taken and each of the lyophilized products of the phosphorylated polypeptide antigen vaccines prepared in Example 1 was added to the system in a ratio of the molar amount of PP-3K to the molar amount of phosphorylated polypeptides of 1:30, and mixed slowly. Then, the reaction system was reacted in a water bath at 25° C. for 30 min. The reaction product was collected, and the mixture was subjected to a desalting treatment with a desalting column from GE Corporation, the system was changed to a PBS solution (pH=7.2), and unbound phosphorylated polypeptides were removed.

(73) Example 5-6: The purified PP-3K solution was quantified by BCA. This protein solution was mixed with a 20 mg/ml dimethylformamide (DMF) solution of maleimide (SMCC) in the ratio of the molar amount of SMCC to the molar amount of PP-3K of 30:1, and the mixture was reacted with slow shaking overnight at 2° C. to 8° C. The mixture was subjected to a desalting treatment with a desalting column from GE Corporation, the system was changed to a PBS solution (pH=7.2-7.4), and unbound SMCC was removed. The reaction product was taken and each of the lyophilized products of the phosphorylated polypeptide antigen vaccines prepared in Example 1 was added to the system in the ratio of the molar amount of PP-3K to the molar amount of phosphorylated polypeptides of 1:50, and mixed slowly. The mixture was reacted with slow shaking overnight at 2° C. to 8° C. The reaction product was collected, and the mixture was subjected to a desalting treatment with a desalting column from GE Corporation, the system was changed to a PBS solution (pH=7.2), and unbound phosphorylated polypeptides were removed.

(74) Example 5-7: The purified PP-3K solution was quantified by BCA. This protein solution was mixed with a 20 mg/ml dimethylformamide (DMF) solution of maleimide (SMCC) in the ratio of the molar amount of SMCC to the molar amount of PP-3K of 5:1, and the mixture was reacted in a water bath at 25° C. for 30 min. The mixture was subjected to a desalting treatment with a dialysis bag having a molecular weight cut-off of 20 kDa (the dialysis volume ratio is 1,000:1), and was dialyzed overnight at 2° C. to 8° C., the system was changed to a PBS solution (pH=7.2-7.4), and unbound SMCC was removed. The reaction product was taken and each of the lyophilized products of the phosphorylated polypeptide antigen vaccines prepared in Example 1 was added to the system in the ratio of the molar amount of PP-3K to the molar amount of phosphorylated polypeptides of 1:100, and mixed slowly. The mixture was reacted with slow shaking overnight at 2° C. to 8° C. The reaction product was collected, and the mixture was subjected to a desalting treatment with a dialysis bag having a molecular weight cut-off of 20 kDa (the dialysis volume ratio is 1,000:1), and was dialyzed overnight at 2° C. to 8° C., the system was changed to a PBS solution (pH=7.2), and unbound phosphorylated polypeptides were removed.

Example 6: Determination of the Ligation Efficiency of the Phosphorylated Polypeptide Antigen Product Coupled with PP-3C

(75) A 10 mM dithiothreitol solution was added to the phosphorylated polypeptide antigen product coupled with PP-3C at a volume ratio of 1:1, and mixed homogeneously, and the mixture was left to stand at room temperature and react for 16 hours. The reaction product was centrifuged at 16,000 g at 4° C. for 15 min. The supernatants were taken for HPLC determination. Using 0.5 mg/mL, 0.25 mg/mL, 0.125 mg/mL, 0.0625 mg/mL, 0.03125 mg/mL, 0.03125 mg/mL, and 0.015625 mg/mL phosphorylated polypeptide dissolved in 10 mM of a dithiothreitol solution as standard, a standard curve of the concentration of the standard versus the peak area of the phosphorylated polypeptide was plotted for quantitatively analyzing the concentration of the test sample (the results are shown in FIGS. 13A, 13B and 13D).

Example 7: Determination of the Ligation Efficiency of the Phosphorylated Polypeptide Antigen Product Coupled with PP-3K

(76) The phosphorylated polypeptide antigen product coupled with PP-3K was subjected to a mass spectrometric analysis, and the concentration of the test sample was quantitatively analyzed using

the phosphorylated polypeptide dissolved in PBS as standard (the results are not shown).

Example 8: Analysis of the Immunogenicity of Phosphorylated Polypeptide Antigen Vaccines in Wild Mice

(77) The phosphorylated polypeptide antigens with sequences as represented by SEQ ID NO: 201, SEQ ID NO: 225, SEQ ID NO: 306, SEQ ID NO: 387, SEQ ID NO: 468, SEQ ID NO: 558, SEQ ID NO: 567, SEQ ID NO: 769, SEQ ID NO: 784, SEQ ID NO: 875, SEQ ID NO: 1020, SEQ ID NO: 1101, SEQ ID NO: 1182, SEQ ID NO: 1272, SEQ ID NO: 1313 and SEQ ID NO: 1330 were dissolved to 1 mg/mL, and mixed with a complete Freund's adjuvant or an incomplete Freund's adjuvant at a volume ratio of 1:1 to form a water-in-oil emulsion so as to prepare vaccines A1 to A16. Each wild mouse was injected intramuscularly with 50 μ L. Each wild mouse was vaccinated four times at two-week intervals using complete Freund's adjuvant for the first immunization and incomplete Freund's adjuvant for the second, third and fourth immunizations. Two weeks after the fourth injection, the mice were subjected to blood sampling to obtain mouse serum for ELISA analysis.

(78) The polypeptide (whose sequence is represented by SEQ ID NO: 1346) which was coupled with BSA and phosphorylated at a position corresponding to amino acid 18 of full-length Tau protein was used as the coating antigen to detect the antiserum titer against a phosphorylation Y18 site; the phosphorylated polypeptide (whose sequence is represented by SEQ ID NO: 1347) which was coupled with BSA and phosphorylated at positions corresponding to amino acids 202 and 205 of full-length Tau protein was used as the coating antigen to detect the antiserum titer against phosphorylation S202 and T205 sites; the phosphorylated polypeptide (whose sequence is represented by SEQ ID NO: 1348) which was coupled with BSA and phosphorylated at a position corresponding to amino acid 212/214 of full-length Tau protein was used as the coating antigen to detect the antiserum titer against phosphorylation T212 and S214 sites; the phosphorylated polypeptide (whose sequence is represented by SEQ ID NO: 1349) which was coupled with BSA and phosphorylated at a position corresponding to amino acid 231/235 of full-length Tau protein was used as the coating antigen to detect the antiserum titer against phosphorylation T231 and S235 sites; the phosphorylated polypeptide (whose sequence is represented by SEQ ID NO: 1350) which was coupled with BSA and phosphorylated at a position corresponding to amino acid 238 of full-length Tau protein was used as the coating antigen to detect the antiserum titer against a phosphorylation S238 site; the phosphorylated polypeptide (whose sequence is represented by SEQ ID NO: 1351) which was coupled with BSA and phosphorylated at a position corresponding to amino acid 262 of full-length Tau protein was used as the coating antigen to detect the antiserum titer against a phosphorylation S262 site; the phosphorylated polypeptide (whose sequence is represented by SEQ ID NO: 1352) which was coupled with BSA and phosphorylated at a position corresponding to amino acid 396 of full-length Tau protein was used as the coating antigen to detect the antiserum titer against a phosphorylation S396 site; the phosphorylated polypeptide (whose sequence is represented by SEQ ID NO: 1353) which was coupled with BSA and phosphorylated at a position corresponding to amino acid 404 of full-length Tau protein was used as the coating antigen to detect the antiserum titer against a phosphorylated S404 site.

(79) The above antigens were respectively dissolved in carbonate buffer with pH=9.5, diluted to 1 μ g/mL, and added to an ELISA plate (100 μ L/well) at 4° C. overnight. The liquid in the plate was discarded, and after the 96-well plate was washed three times with PBST (300 μ L/well), the liquid in the plate was discarded. A 3% BSA (PBS) solution was added (200 μ L/well), and incubation was carried out at 37° C. for 1 hour. The liquid in the plate was discarded, and after the 96-well plate was washed three times with PBST (300 μ L/well), the liquid in the plate was discarded. Mouse serum diluted by gradient with a 1% BSA (PBS) solution was added (100 μ L/well), and incubation was carried out at 37° C. for 2 hour. The liquid in the plate was discarded, and after the 96-well plate was washed three times with PBST (300 L/well), the liquid in the plate was discarded. A goat-anti-mouse HRP labeled antibody solution diluted with a 1% BSA (PBS) solution was added (100

μL/well), and incubation was carried out at 37° C. for 1 hour. The liquid in the plate was discarded, and after the 96-well plate was washed three times with PBST (300 μL/well), the liquid in the plate was discarded. TMB was added (300 μL/well) and incubation was carried out in the dark at room temperature for 25 min. 1M sulfuric acid aqueous solution was added to terminate the reaction. Absorbance was detected at 450 nm using a microplate reader

(80) The levels of the antibodies against different antigens in the serum after the fourth immunization by vaccines A1 to A16 are shown in FIG. 6. The results show that all mice in groups of vaccines A1 to A6 can successfully produce antibodies against the pY18 site after immunization, wherein vaccines A2 and A5 can produce high concentrations of antibodies; and vaccines A7 to A16 can hardly produce antibodies against the pY18 site (FIG. 6A). All the groups of mice immunized with vaccines A1 to A2 and A7 to A11 can successfully produce antibodies against the pS202/pT205 site after immunization, wherein vaccines A8 to A11 can produce high concentrations of antibodies; and vaccines A3 to A6 and A12 to A16 can hardly produce antibodies against the pS202/pT205 site (FIG. 6B). All the groups of mice immunized with vaccines A3 and A12 to A14 can successfully produce high concentrations of antibodies against the pT212/pS214 site after immunization, vaccines A6, A8 and A11 can produce low concentrations of antibodies; and vaccines A1, A2, A4, A5, A7, A9 to A11, A15 and A16 can hardly produce antibodies against the pT212/pS214 site (FIG. 6C). All the groups of mice immunized with vaccines A4, A7 to A8 and A12 can successfully produce high concentrations of antibodies against the pS231/pS235 site after immunization, vaccines A1, A5 and A14 can produce low concentrations of antibodies; and vaccines A2, A3, A6, A9 to A11, A13, A15 and A16 can hardly produce antibodies against the pS231/pS235 site (FIG. 6D). All the groups of mice immunized with vaccines A5, A9, A13 and A15 to A16 can successfully produce high concentrations of antibodies against the pS238 site after immunization, vaccines A8 and A14 can also produce relatively high concentrations of antibodies; and vaccines A1 to A4, A6, A7 and A10 to A12 can hardly produce antibodies against the pS238 site (FIG. 6E). All the groups of mice immunized with vaccines A5, A9, A13 and A16 can successfully produce high concentrations of antibodies against the pS262 site after immunization, vaccines A7 and A14 can also produce relatively high concentrations of antibodies; and vaccines A1 to A4, A6, A8 and A10 to A12 can hardly produce antibodies against the pS238 site (FIG. 6F). All the groups of mice immunized with vaccines A6, A10, A11 and A14 to A16 can successfully produce high concentrations of antibodies against the pS396 site after immunization, vaccine A8 can produce low concentrations of antibodies; and vaccines A1 to A5, A7, A9 and A12 to A13 can hardly produce antibodies against the pS396 site (FIG. 6G). All the groups of mice immunized with vaccine A6, A10, A11 and A14 to A16 can successfully produce high concentrations of antibodies against the pS404 site after immunization, vaccine A8 can produce low concentrations of antibodies; and vaccine A1 to A5, A7, A9 and A12 to A13 can hardly produce antibodies against the pS396 site (FIG. 6H). In conclusion, after used for immunizing animals, all the phosphorylated polypeptide vaccines designed in Example 1 can produce antibodies against the corresponding phosphorylation sites and have immunogenicity, and vaccine A8 and vaccine A14 can produce cross-protection responses against other phosphorylation sites; meanwhile, it was found that immunization with relatively long phosphorylated peptides can lead to relatively high concentrations of antibodies.

Example 9: Evaluation of the Immunogenicity of Phosphorylated Polypeptide Antigens Coupled with the Norovirus P Protein PP-3C in Wild Mice Immunized with the Same by Intramuscular Injection

(81) The PP-3C protein was coupled with A1, A5, A9, A10, A11, A12 and A15 respectively to prepare complex vaccines, and wild mice were immunized with the complex vaccines by intramuscular injection. Each group contained 6 mice. The immune dosage of the protein was 25 μg/animal. The immunization was carried out at week 0, week 2, week 4 and week 6. Blood was taken two weeks after every immunization to obtain serum for ELISA assessment. The assessment

was performed by the same method as in Example 8. The results are shown in FIGS. 7A-7G. As apparent from the figures, mice immunized with complex vaccine A1 can produce specific antibodies against pY18 and pS202/pT205 (FIG. 7A); mice immunized with complex vaccine A5 can produce specific antibodies against pY18, pS238 and pS262 (FIG. 7B); mice immunized with complex vaccine A9 can produce specific antibodies against pS202/pT205, pS238 and pS262 (FIG. 7C); mice immunized with complex vaccine A9 can produce specific antibodies against pS202/pT205, pS396 and pS404 (FIG. 7D); mice immunized with complex vaccine A11 can produce specific antibodies against pS202/pT205, pS396 and pS404 (FIG. 7E); mice immunized with complex vaccine A12 can produce specific antibodies against pT212/pS214 and pS231/pS235 (FIG. 7F); and mice immunized with complex vaccine A15 can produce specific antibodies against pS238, pS262, pS396 and pS404 (FIG. 7G). Thus, it can be seen that the concentration of the specific antibodies in the serum increased with each intramuscular immunization with the complex vaccine in each group, and after the fourth immunization high concentrations of antibodies were produced, wherein pY18 and pS202/pT205 epitopes are prone to leading to high concentrations of specific antibodies. Compared with other immunization routes, intramuscular immunization generally can lead to higher concentrations of antibodies.

Example 10: Evaluation of the Immunogenicity of Phosphorylated Polypeptide Antigens Coupled with the Norovirus P Protein PP-3C in Wild Mice Immunized with the Same by Intraperitoneal Injection

(82) The PP-3C protein was coupled with A2, A3, A7, A11 and A14 respectively to prepare complex vaccines, and wild mice were immunized with the complex vaccines by intraperitoneal injection. Each group contained 6 mice. The immune dosage of the protein was 25 $\mu\text{g}/\text{animal}$. The immunization was carried out at week 0, week 2, week 4 and week 6. Blood was taken two weeks after every immunization to obtain serum for ELISA assessment. The assessment was performed by the same method as in Example 10. The results were shown in FIGS. 7H-7M. As apparent from the figures, mice immunized with complex vaccine A2 can produce specific antibodies against pY18 and pS202/pT205 (FIG. 7H); mice immunized with complex vaccine A3 can produce specific antibodies against pY18 and pT212/pS214 (FIG. 7J); mice immunized with complex vaccine A7 can produce specific antibodies against pS202/pT205 and pS231/pS235 (FIG. 7K); mice immunized with complex vaccine A11 can produce specific antibodies against pS202/pT205, pS396 and pS404 (FIG. 7L); mice immunized with complex vaccine A14 can produce specific antibodies against pT212/pS214, pS396 and pS404 (FIG. 7M). Thus, it can be seen that the concentration of the specific antibodies in the serum increased with each intraperitoneal immunization with the complex vaccine in each group, and after the fourth immunization high concentrations of antibodies were produced.

Example 11: Evaluation of the Immunogenicity of Phosphorylated Polypeptide Antigens Coupled with the Norovirus P Protein PP-3C in Wild Mice Immunized with the Same by Subcutaneous Injection

(83) The PP-3C protein was coupled with A4, A6, A8, A11, A13 and A16 respectively to prepare complex vaccines, and wild mice were immunized with the complex vaccines by subcutaneous injection. Each group contained 6 mice. The immune dosage of the protein was 25 $\mu\text{g}/\text{animal}$. The immunization was carried out at week 0, week 2, week 4 and week 6. Blood was taken two weeks after every immunization to obtain serum for ELISA assessment. The assessment was performed by the same method as in Example 10. The results were shown in FIGS. 7N-7S. As apparent from the figures, mice immunized with complex vaccine A4 can produce specific antibodies against pY18 and pS231/pS235 (FIG. 7N); mice immunized with complex vaccine A6 can produce specific antibodies against pY18, pS396 and pS404 (FIG. 7O); mice immunized with complex vaccine A8 can produce specific antibodies against pS202/pT205 and pS231/pS235 (FIG. 7P); mice immunized with complex vaccine A11 can produce specific antibodies against pS202/pT205, pS396 and pS404 (FIG. 7Q); mice immunized with complex vaccine A13 can produce specific

antibodies against pT212/pS214, pS396 and pS404 (FIG. 7R); and mice immunized with complex vaccine A16 can produce specific antibodies against pS238, pS262, pS396 and pS404 (FIG. 7S). Thus, it can be seen that the concentration of the specific antibodies in the serum increased with each subcutaneous immunization with the complex vaccine in each group, and after the fourth immunization high concentrations of antibodies were produced.

Example 12: Evaluation of the Immunogenicity of Phosphorylated Polypeptide Antigens Coupled with a Norovirus P Protein in Wild Mice

(84) Take vaccine A11 as an example. Wild mice were intramuscularly immunized with A11 coupled with PP-3C or PP-3K and combined with CpG, AS02, AS03, MF59 and AS02+CpG adjuvants respectively. Each group contained 6 mice. The immune dosage of the protein was 25 µg/animal. The immunization was carried out at week 0, week 2, week 4, week 6 and week 12. Blood was taken two weeks after every immunization to obtain serum for ELISA assessment. In the ELISA experiments, the phosphorylated polypeptide (whose sequence is SEQ ID NO: 1354) coupled with BSA was used as the coating antigen, and was phosphorylated at sites corresponding to amino acids 202, 205, 396 and 404 of full-length Tau protein. A standard curve was plotted with the concentration of the AT8 antibody (Thermo scientific MN1020) diluted by gradient versus the O.D. value to calibrate the concentration of the antibodies in the mouse serum. The change in antibody concentration in each group is shown in FIG. 8. As can be seen from FIG. 8, vaccine A11 coupled with a norovirus P protein and combined with different adjuvants could elicit relatively high immune responses in wild mice, wherein the vaccine in combination with AS02, AS03 and MF59 enabled gradual production of relatively high concentrations of antibodies through four consecutive immunizations, and failed to stimulate and elicit a relatively high immune response in one immunization after an interval of 6 weeks; the vaccine combined with CpG adjuvant and AS02+CpG adjuvant enabled gradual production of relatively low concentrations of antibodies through four consecutive immunizations, and stimulated the production of a relatively high concentration of antibodies in one immunization after an interval of 6 weeks, and the antibody content could maintain a high level for a long time. The production of antibodies in mice immunized with vaccine A11 coupled with PP-3C was basically the same with that in mice immunized with vaccine A11 coupled with PP-3K. Immunization of wild mice with the other phosphorylated polypeptide vaccines produced similar effects (the results are not shown).

Example 13: Evaluation of the Immunogenicity of Phosphorylated Polypeptide Antigen A5 Coupled with a Norovirus P Protein in Wild Mice

(85) Wild mice were intramuscularly immunized with the vaccine prepared by combining phosphorylated polypeptide antigen A5 coupled with PP-3K with AS02+CpG adjuvant. Each group contained 6 mice. The immune dosage of the protein was 25 µg/animal. The immunization was carried out at week 0, week 2, week 4, week 6 and week 12. Blood was taken two weeks after every immunization to obtain serum for ELISA assessment. In the ELISA experiments, the phosphorylated polypeptide (whose sequence is SEQ ID NO: 1355) coupled with BSA was used as the coating antigen, and was phosphorylated at sites corresponding to amino acids 18, 238 and 262 of full-length Tau protein. The change in the concentration of antibodies in the mouse serum is shown in FIG. 9A. The results showed that intramuscular immunization with the PP-3K-A5 complex vaccine in combination with AS02+CpG adjuvant led to an increase in antibodies with each immunization, the vaccine had good immunogenicity, and high antibody concentrations could be maintained for at least 6 weeks after the last immunization, with antibody titer $\geq 12,800$.

Example 14: Evaluation of the Immunogenicity of Phosphorylated Polypeptide Antigen A5 Coupled with a Norovirus P Protein in P301S Transgenic Mice

(86) P301S transgenic mice were intramuscularly immunized with the vaccine prepared by combining phosphorylated polypeptide antigen A5 coupled with PP-3K with AS02+CpG adjuvant. Each group contained 8 mice. The immune dosage of the protein was 25 µg/animal. The immunization was carried out at week 0, week 2, week 4, week 6 and week 12. Blood was taken

two weeks after every immunization to obtain serum for ELISA assessment. In the ELISA experiments, the phosphorylated polypeptide (whose sequence is SEQ ID NO: 1355) coupled with BSA was used as the coating antigen, and was phosphorylated at sites corresponding to amino acids 18, 238 and 262 of full-length Tau protein. The change in the concentration of antibodies in the mouse serum is shown in FIG. 9B. The results showed that intramuscular immunization with the PP-3K-A5 complex vaccine in combination with AS02+CpG adjuvant led to an increase in antibodies with each immunization, the vaccine had good immunogenicity, and high antibody concentrations could be maintained for at least 6 weeks after the last immunization, with antibody titer $\geq 12,800$.

Example 15: Evaluation of the Immunogenicity of Phosphorylated Polypeptide Antigen A11 Coupled with a Norovirus P Protein in P301S Transgenic Mice

(87) P301S transgenic mice were intramuscularly immunized with the vaccine prepared by combining phosphorylated polypeptide antigen A11 coupled with PP-3C with AS02+CpG adjuvant. Each group contained 12 mice. The immune dosage of the protein was 25 $\mu\text{g}/\text{animal}$. The immunization was carried out at week 0, week 2, week 4, week 6 and week 12. Blood was taken two weeks after every immunization to obtain serum for ELISA assessment. In the ELISA experiments, the phosphorylated polypeptide (whose sequence is SEQ ID NO: 1354) coupled with BSA was used as the coating antigen, and was phosphorylated at sites corresponding to amino acids 202, 205, 396 and 404 of full-length Tau protein. The change in the concentration of antibodies in the mouse serum is shown in FIG. 9C. The results showed that intramuscular immunization with the PP-3K-A11 complex vaccine in combination with AS02+CpG adjuvant led to an increase in antibodies with each immunization, the vaccine had good immunogenicity, and high antibody concentrations could be maintained for at least 6 weeks after the last immunization, with antibody titer $\geq 12,800$.

Example 16: Evaluation of the Immunogenicity of Phosphorylated Polypeptide Antigen A11 Coupled with a Norovirus P Protein in P301S Transgenic Mice

(88) P301S transgenic mice were intramuscularly immunized with the vaccine prepared by combining phosphorylated polypeptide antigen A11 coupled with PP-3K with AS02+CpG adjuvant. Each group contained 12 mice. The immune dosage of the protein was 25 $\mu\text{g}/\text{animal}$. The immunization was carried out at week 0, week 2, week 4, week 6 and week 12. Blood was taken two weeks after every immunization to obtain serum for ELISA assessment. In the ELISA experiments, the phosphorylated polypeptide (whose sequence is SEQ ID NO: 1354) coupled with BSA was used as the coating antigen, and was phosphorylated at sites corresponding to amino acids 202, 205, 396 and 404 of full-length Tau protein. The change in the concentration of antibodies in the mouse serum is shown in FIG. 9D. The results showed that intramuscular immunization with the PP-3K-A5 complex vaccine in combination with AS02+CpG adjuvant led to an increase in antibodies with each immunization, the vaccine had good immunogenicity, and high antibody concentrations could be maintained for at least 6 weeks after the last immunization, with antibody titer $\geq 12,800$.

Example 17: Evaluation of the Immunogenicity of Phosphorylated Polypeptide Antigen A12 Coupled with a Norovirus P Protein in Wild Mice

(89) Wild mice were intramuscularly immunized with the vaccine prepared by combining phosphorylated polypeptide antigen A12 coupled with PP-3K with AS02+CpG adjuvant. Each group contained 6 mice. The immune dosage of the protein was 25 $\mu\text{g}/\text{animal}$. The immunization was carried out at week 0, week 2, week 4, week 6 and week 12. Blood was taken two weeks after every immunization to obtain serum for ELISA assessment. In the ELISA experiments, the phosphorylated polypeptide (whose sequence is SEQ ID NO: 1356) coupled with BSA was used as the coating antigen, and was phosphorylated at sites corresponding to amino acids 212, 214, 231 and 235 of full-length Tau protein. The change in the concentration of antibodies in the mouse serum is shown in FIG. 9E. The results showed that intramuscular immunization with the PP-3K-

A12 complex vaccine in combination with AS02+CpG adjuvant led to an increase in antibodies with each immunization, the vaccine had good immunogenicity, and high antibody concentrations could be maintained for at least 6 weeks after the last immunization, with antibody titer $\geq 12,800$.

Example 18: Evaluation of the Immunogenicity of Phosphorylated Polypeptide Antigen A12 Coupled with a Norovirus P Protein in P301S Transgenic Mice

(90) P301S transgenic mice were intramuscularly immunized with the vaccine prepared by combining phosphorylated polypeptide antigen A12 coupled with PP-3K with AS02+CpG adjuvant. Each group contained 12 mice. The immune dosage of the protein was 25 $\mu\text{g}/\text{animal}$. The immunization was carried out at week 0, week 2, week 4, week 6 and week 12. Blood was taken two weeks after every immunization to obtain serum for ELISA assessment. In the ELISA experiments, the phosphorylated polypeptide (whose sequence is SEQ ID NO: 1356) coupled with BSA was used as the coating antigen, and was phosphorylated at sites corresponding to amino acids 212, 214, 231 and 235 of full-length Tau protein. The change in the concentration of antibodies in the mouse serum is shown in FIG. 9F. The results showed that intramuscular immunization with the PP-3K-A12 complex vaccine combined with AS02+CpG adjuvant led to an increase in antibodies with each immunization, the vaccine had good immunogenicity, and high antibody concentrations could be maintained for at least 6 weeks after the last immunization, with antibody titer $\geq 12,800$.

Example 19: Evaluation of the Cellular Immunity of Phosphorylated Polypeptide Antigen A5 Coupled with a Norovirus P Protein in P301S Transgenic Mice—ELISPOT Detection

(91) A96-well plate was coated with the monoclonal antibody against cytokine interferon γ (from elispot kit purchased from BD Corporation) in a concentration of 5 $\mu\text{g}/\text{mL}$ (50 $\mu\text{L}/\text{well}$), and covered at 4° C. overnight. After discarding the coating antibody and washing once with a complete medium containing 10% fetal bovine serum, 200 μL of this complete medium was added to each well. Blocking was carried out at 37° C. for 1 hour, and then the medium was discarded. Mice used in the experiment were sacrificed by neck-pulling. Spleen cells of the mice were taken out and formulated into a cell suspension in a cell concentration of $10^7/\text{mL}$. The cell suspension was added to the coated 96-well plate (100 $\mu\text{L}/\text{well}$). 100 μL of 1 $\mu\text{g}/\text{mL}$ prokaryotically expressed full-length Tau protein and nonphosphorylated A5 polypeptide antigen were added to each well. The plate was then cultured at 37° C. in an incubator containing 5% CO.sub.2 for 24 h to stimulate and activate the cells. 24 h later, the plate was washed two times with sterile water, and six times with sterile PBST (pH7.4, 0.01 mol/L PBS, containing 0.05% Tween-20) buffer to wash the cells away. 50 μL of 2 $\mu\text{g}/\text{mL}$ antibody against interferon γ (from elispot kit purchased from BD Corporation) was added to each well and incubated at room temperature for 2 hours. The 96-well plate was washed, and horse radish peroxidase labeled biotin secondary antibody (from elispot kit purchased from BD Corporation) was added (50 $\mu\text{L}/\text{well}$). The plate was cultured at room temperature for 2 h, and washed four times with PBST, and two times with PBS. 50 μL of Elispot color developing solution (AEC substrate) was added to each well and reacted in the dark at room temperature for 5-60 min. The staining solution was discarded, and the plate was washed with distilled water. After being dried overnight, the sample was calculated for the number of activated cells using a microscope.

(92) The experimental results are shown in FIGS. 10A and 10B. The PP-3K-A5 complex vaccine combined with AS02+CpG adjuvant could stimulate spleen cells to produce fewer spots after five immunizations, which demonstrates no evident T cell response occurring in the body, and as described in Example 14, this immunization strategy can stimulate the mice to produce the highest titer of specific antibodies against phosphorylated A5 polypeptide. Considering the safety of vaccines, this immunization strategy can be selected as a preferred immunization strategy.

Example 20: Evaluation of the Cellular Immunity of Phosphorylated Polypeptide Antigen A11 Coupled with a Norovirus P Protein in P301S Transgenic Mice—ELISPOT Detection

(93) The experiment was performed by the same method as in Example 19. The stimulus was 100

μL of 1 μg/mL prokaryotically expressed full-length Tau protein and nonphosphorylated A11 polypeptide antigen.

(94) The experimental results are shown in FIGS. **10C** and **10D**. The PP-3K-A11 complex vaccine combined with AS02+CpG adjuvant could stimulate spleen cells to produce fewer spots after five immunizations, which demonstrates no evident T cell response occurring in the body, and as described in Example 15, this immunization strategy can stimulate the mice to produce the highest titer of specific antibodies against phosphorylated A11 polypeptide. Considering the safety of vaccines, this immunization strategy can be selected as a preferred immunization strategy.

Example 21: Evaluation of the Cellular Immunity of Phosphorylated Polypeptide Antigen A11 Coupled with a Norovirus P Protein in P301S Transgenic Mice—ELISPOT Detection

(95) The experiment was performed by the same method as in Example 19. The stimulus was 100 μL of 1 μg/mL prokaryotically expressed full-length Tau protein and nonphosphorylated A11 polypeptide antigen.

(96) The experimental results are shown in FIGS. **10E** and **10F**. The PP-3K-A11 complex vaccine combined with AS02+CpG adjuvant could stimulate spleen cells to produce fewer spots after five immunizations, which demonstrates no evident T cell response occurring in the body, and as described in Example 15, this immunization strategy can stimulate the mice to produce the highest titer of specific antibodies against phosphorylated A11 polypeptide. Considering the safety of vaccines, this immunization strategy can be selected as a preferred immunization strategy.

Example 22: Evaluation of the Cellular Immunity of Phosphorylated Polypeptide Antigen A12 Coupled with a Norovirus P Protein in P301S Transgenic Mice—ELISPOT Detection

(97) The experiment was performed by the same method as in Example 19. The stimulus was 100 μL of 1 μg/mL prokaryotically expressed full-length Tau protein and nonphosphorylated A12 polypeptide antigen.

(98) The experimental results are shown in FIGS. **10G** and **10H**. The PP-3K-A12 complex vaccine combined with AS02+CpG adjuvant could stimulate spleen cells to produce fewer spots after five immunizations, which demonstrates no evident T cell response occurring in the body, and as described in Example 15, this immunization strategy can stimulate the mice to produce the highest titer of specific antibodies against phosphorylated A12 polypeptide. Considering the safety of vaccines, this immunization strategy can be selected as a preferred immunization strategy.

Example 23: Evaluation of the Behavior of P301S Transgenic Mice Immunized with Phosphorylated Polypeptide Antigen A5 Coupled with a Norovirus P Protein

(99) P301S transgenic mice were immunized with phosphorylated polypeptide antigen A5 which was coupled with PP-3K and combined with AS02+CpG adjuvant. Eight weeks after the start of the experiment, mice in the immune group were subjected to a rota-rod endurance test every two weeks. The mice were placed on the mouse rota-rod apparatus, and after the rota-rod apparatus was turned on, its rotation speed rose from 5 rpm to 40 rpm within 1.5 minutes and remained constant. The time until the mice fell from the rota-rod within 300 s was recorded. The experiment showed that the immunized mice had stronger endurance in the rota-rod experiment than the PBS group (the results are shown in FIG. **11A**).

Example 24: Evaluation of the Behavior of P301S Transgenic Mice Immunized with Phosphorylated Polypeptide Antigen A11 Coupled with a Norovirus P Protein

(100) P301S transgenic mice were immunized with phosphorylated polypeptide antigen A11 which was coupled with PP-3C and combined with AS02+CpG adjuvant. Eight weeks after the start of the experiment, mice in the immune group were subjected to a rota-rod endurance test every two weeks. The mice were placed on the mouse rota-rod apparatus, and after the rota-rod apparatus was turned on, its rotation speed rose from 5 rpm to 40 rpm within 1.5 minutes and remained constant. The time until the mice fell from the rota-rod within 300 s was recorded. The experiment showed that the immunized mice had stronger endurance in the rota-rod experiment than the PBS group (the results are shown in FIG. **11B**).

Example 25: Evaluation of the Behavior of P301S Transgenic Mice Immunized with

Phosphorylated Polypeptide Antigen A11 Coupled with a Norovirus P Protein

(101) P301S transgenic mice were immunized with phosphorylated polypeptide antigen A11 which was coupled with PP-3K and combined with AS02+CpG adjuvant. Eight weeks after the start of the experiment, mice in the immune group were subjected to a rota-rod endurance test every two weeks. The mice were placed on the mouse rota-rod apparatus, and after the rota-rod apparatus was turned on, its rotation speed rose from 5 rpm to 40 rpm within 1.5 minutes and remained constant. The time until the mice fell from the rota-rod within 300 s was recorded. The experiment showed that the immunized mice had stronger endurance in the rota-rod experiment than the PBS group (the results are shown in FIG. 11C).

Example 26: Evaluation of the Behavior of P301S Transgenic Mice Immunized with

Phosphorylated Polypeptide Antigen A12 Coupled with a Norovirus P Protein

(102) P301S transgenic mice were immunized with phosphorylated polypeptide antigen A12 which was coupled with PP-3K and combined with AS02+CpG adjuvant. Eight weeks after the start of the experiment, mice in the immune group were subjected to a rota-rod endurance test every two weeks. The mice were placed on the mouse rota-rod apparatus, and after the rota-rod apparatus was turned on, its rotation speed rose from 5 rpm to 40 rpm within 1.5 minutes and remained constant. The time until the mice fell from the rota-rod within 300 s was recorded. The experiment showed that the immunized mice had stronger endurance in the rota-rod experiment than the PBS group (the results are shown in FIG. 11D).

Example 27: Evaluation of the Behavior of P301S Transgenic Mice Immunized with

Phosphorylated Polypeptide Antigen A5 Coupled with a Norovirus P Protein

(103) P301S transgenic mice were immunized with phosphorylated polypeptide antigen A5 coupled to PP-3K and combined with AS02+CpG adjuvant. Ten weeks after the start of the experiment, a nesting test was performed. All mice were transferred to new sterile mouse boxes. New sterile bedding was spread evenly to a thickness of 0.8 cm/box. Two 5 cm×5 cm cotton pieces were placed in the center of the right side of each mouse box, the total weight of the cotton being 1.2 g to 1.3 g. The mice were transferred to the mouse boxes at 16:00 and the indoor lighting was turned off. The lighting was turned on at 7:00 the next day and the nesting effect of mice was scored at 8:00. The labels with mouse numbers were hidden and three trained operators performed the scoring independently of each other. The average of the three scores represents the score of one mouse in the nesting test.

(104) The scoring criteria were as follows: 1: the cotton was not completely torn; 2: the cotton was completely torn for nest building, the nest being shallow and flat; 3: the cotton was completely torn for nest building, but the nest had other notches besides those for exposing drinking water and the like, or the upper edge of the nest entrance was lower than the mouse's head; 4: the cotton was completely torn for nest building, and the nest entrance was higher than the mouse's head, or the nest was three-dimensional and had no notch.

(105) The nesting test can reflect the integrity of the consciousness of mice. The experiment showed that the immunized mice performed significantly better than the PBS group in the nesting test (the results are shown in FIG. 12A).

Example 28: Evaluation of the Behavior of P301S Transgenic Mice Immunized with

Phosphorylated Polypeptide Antigen A11 Coupled with a Norovirus P Protein

(106) P301S transgenic mice were immunized with phosphorylated polypeptide antigen A11 coupled with PP-3C and combined with AS02+CpG adjuvant. Ten weeks after the start of the experiment, a nesting test was performed by the same method as in Example 27. The experiment showed that the immunized mice performed significantly better than the PBS group in the nesting test (the results are shown in FIG. 12B).

Example 29: Evaluation of the Behavior of P301S Transgenic Mice Immunized with

Phosphorylated Polypeptide Antigen A11 Coupled with a Norovirus P Protein

(107) P301S transgenic mice were immunized with phosphorylated polypeptide antigen A11 coupled with PP-3K and combined with AS02+CpG adjuvant. Ten weeks after the start of the experiment, a nesting test was performed by the same method as in Example 27. The experiment showed that the immunized mice performed significantly better than the PBS group in the nesting test (the results are shown in FIG. 12C).

Example 30: Evaluation of the Behavior of P301S Transgenic Mice Immunized with Phosphorylated Polypeptide Antigen A12 Coupled with a Norovirus P Protein

(108) P301S transgenic mice were immunized with phosphorylated polypeptide antigen A12 coupled with PP-3K and combined with AS02+CpG adjuvant. Ten weeks after the start of the experiment, a nesting test was performed by the same method as in Example 27. The experiment showed that the immunized mice performed significantly better than the PBS group in the nesting test (the results are shown in FIG. 12D).

Example 31: Construction of pET28a-C-PA Plasmid

(109) The plasmid containing C-PA was obtained through sequence optimization and gene synthesis (GL Bio).

(110) The C-PA genes were enriched by PCR method and the sequence of the genes was represented by SEQ ID NO: 1363. The specific method was as follows:

(111) TABLE-US-00010 SEQ ID NO: 1365 (forward):

TTTAAC TTAAAGAAGGAGATATACATATGGGTGGTGGTGGTTCTTGCGG CGGCG, and
SEQ ID NO: 1366 (reverse):

GTGGTGGTGGTGGTGGTGGTCTCGAGTTATTTAATACGCAGATACTGGCCA ATCA;

(112) The plasmid containing C-PA was subjected to PCR reaction, wherein the PCR reaction system was a KOD-Plus DNA polymerase system (purchased from TOYOBO Corporation), and the total volume of the reaction system was 50 μ L (5 μ L buffer, 0.2 mM dNTP, 1 mM magnesium sulfate, 0.3 μ M upstream primer and 0.3 μ M downstream primer, 50 ng template DNA, 1 μ L KOD enzyme, and water which was added to a final volume of 50 μ L). PCR was carried out in accordance with the reaction system instructions to obtain 50 μ L of PCR products. The products were subjected to agarose gel electrophoresis, and the fragments of interest were recovered using gel recovery kit (purchased from Tiangen Biotech Co., Ltd.)

(113) 1 μ L of BamHI enzyme and 1 μ L of XhoI enzyme (purchased from Takara Corporation) and 5 μ L of enzyme digestion buffer (purchased from Takara Corporation) were added respectively to 2 μ g of pET28a vector plasmid (purchased from Novagen Corporation), and finally sterile water was added to the system to reach a final volume of 50 μ L. Digestion was carried out at 37° C. for 2 hours. The products were subjected to agarose gel electrophoresis, and the vector fragments were recovered using gel recovery kit (purchased from Tiangen Biotech Co., Ltd.) to obtain a linear plasmid vector.

(114) The above double enzyme-digested vector fragment and fragment of interest (with a molar ratio of 1:3, and a total volume of 5 μ L) were mixed, and 5 μ L of homologous recombination mix (purchased from Taihe Corporation) was added. The ligation was carried out at 25° C. for 15 min. 10 μ L of the ligated products were added to DH10B competent cells (purchased from Taihe Corporation), and placed on ice for 30 min. After that, the cells were subjected to heat shock at 42° C. for 30 s, placed on ice for 2 min, and then were added to 600 L of fluid LB medium without resistance, and recovered at 200 rpm at 37° C. for 1 h. The culture broth was plated on a LB solid culture plate containing kanamycin (15 μ g/mL), and were placed upside down at 37° C. overnight. The recombinant clones were obtained, and the sequence was verified by sequencing, which was represented by SEQ ID NO: 1363. A pET28a-C-PA plasmid that can express C-PA protein was obtained, as shown in FIG. 15A.

Example 32: Expression and Purification of a C-PA Protein

(115) 32.1 Expression of a C-PA Protein

(116) 1 μ L of the recombinant plasmids prepared in the above examples were respectively added to

100 μ L of *E. coli* BL21 competent cells (purchased from TransGen Corporation), ice-bathed for 30 min, heat shocked for 90 s in a water-bath at 42° C., and then ice-bathed for 2 min. 600 μ L of LB medium was added to the mixture, and cultured at 180 rpm/min at 37° C. for 1 h. The mixture was coated evenly on a LB solid medium containing kanamycin (15 μ g/mL) resistance and cultured at 37° C. for 24 h to obtain strains that can stably express recombinant proteins. A growing colony was picked and inoculated into 20 mL of LB medium. The mixture was cultured at 220 rpm at 37° C. When the OD value of the culture mixture reached 2.0, induction by Isopropyl β -D-Thiogalactoside (IPTG at a final concentration of 1.0 mmol/L) was carried out at 220 rpm at 37° C. for six hours. After the induction, the culture broth was centrifuged at 4,000 rpm for 20 min. The supernatant was discarded, and the bacterial precipitates were resuspended with PBS. Centrifugation was conducted again at 4,000 rpm for 20 min and the supernatant was discarded to obtain the bacterial precipitates containing proteins of interest.

(117) 32.2 Purification of a C-PA Protein

(118) The bacterial precipitates obtained in 32.1 were resuspended by adding 20 mL of protein buffer (pH=7.0, containing 50 mM Tris and 300 mM KCl). The bacteria were lysed by sonication on ice for 30 min. The mixture was centrifuged at 12,000 rpm at 4° C. for 30 min and the supernatant was discarded. The collected precipitates were resuspended in 8M urea buffer (pH=7.0, containing 50 mM Tris, 300 mM KCl and 8M urea) and stirred overnight at 4° C. The collected solution was centrifuged at 16,000 rpm at 4° C. for 30 min. The supernatant was collected and transferred to a dialysis bag with a cut-off size of 10 kd. Dialysis was performed in the ratio of the volume of glycine dialysate to the volume of protein solution of 10:1, and the fluid was changed every 4 hours. The dialysis was performed 6 times to slowly renature the protein. The collected protein solution was concentrated by ultrafiltration to a protein concentration of 1.0 mg/mL to 2.0 mg/mL. The results of SDS-PAGE of the protein are shown in FIG. 15B, and the molecular weight of the protein is about 24 kd.

Example 33: Coupling Reaction of Phosphorylated Polypeptide Antigen Vaccines with BLP

Example 33.1: Ligation Reaction of C-PA with BLP

(119) The method for preparing BLP was described in the Chinese Patent Application CN 105968213. 1 mL of 15 mg/mL BLP solution was centrifuged at 5,000 g at 4° C. for 30 min, and the supernatant was discarded. BLP was resuspended with 5 mL of C-PA solution and slowly shaken at 25° C. for 30 min to prepare C-PA-BLP. Centrifuge was performed at 5,000 g for 30 min at 4° C., the supernatant was discarded, and resuspension was performed with 10 mL sterile PBS (pH=7.2). This operation was repeated 4 times. The washed precipitates were resuspended with 1 mL PBS, OD600 was detected, and the concentration of C-PA-BLP was calibrated with a BLP standard.

Example 33.2: Ligation Reaction of C-PA with BLP

(120) The method for preparing BLP was described in the Chinese Patent Application CN 105968213. 1 mL of 15 mg/mL BLP solution was centrifuged at 5,000 g at 4° C. for 30 min, and the supernatant was discarded. BLP was resuspended with 3 mL of C-PA solution and slowly shaken at 4° C. for 6 h to prepare C-PA-BLP. Centrifuge was performed at 5,000 g for 30 min at 4° C., the supernatant was discarded, and resuspension was performed with 10 mL sterile PBS (pH=7.2). This operation was repeated 4 times. The washed precipitates were resuspended with 1 mL PBS, OD600 was detected, and the concentration of C-PA-BLP was calibrated with a BLP standard.

Example 33.3: Coupling Reaction of the Phosphorylated Polypeptide Antigen Vaccines with C-PA-BLP

(121) The phosphorylated polypeptide in each of the phosphorylated polypeptide antigen vaccines in lyophilized form prepared in Example 1 was diluted with a 0.1 M ammonium bicarbonate solution (pH=7.5) to a concentration of 1 mg/mL and was quantified. In the ratio of the molar amount of the C-PA protein to the molar amount of the polypeptide of 1:30, the polypeptide

solution was slowly added to the precipitates obtained after centrifugation of C-PA-BLP, and mixed slowly. The mixed system was incubated at 2° C. to 8° C. with slow shaking for 48 hours. The obtained product was centrifuged and resuspended with PBS to remove unbound polypeptide.

Example 33.4: Coupling Reaction of the Phosphorylated Polypeptide Antigen Vaccines with C-PA-BLP

(122) The phosphorylated polypeptide in each of the phosphorylated polypeptide antigen vaccines in lyophilized form prepared in Example 1 was diluted with a 0.1 M ammonium bicarbonate solution (pH=8.0) to a concentration of 2 mg/mL and was quantified. In the ratio of the molar amount of the C-PA protein to the molar amount of the polypeptide of 1:100, the polypeptide solution was slowly added to the precipitates obtained after centrifugation of C-PA-BLP, and mixed slowly. The mixed system was incubated at 2° C. to 8° C. with slow shaking for 48 hours. The obtained product was centrifuged and resuspended with PBS to remove unbound polypeptide.

Example 33.5: Coupling Reaction of the Phosphorylated Polypeptide Antigen Vaccines with C-PA-BLP

(123) The phosphorylated polypeptide in each of the phosphorylated polypeptide antigen vaccines in lyophilized form prepared in Example 1 was diluted with a 0.1 M ammonium bicarbonate solution (pH=8.0) to a concentration of 2 mg/mL and was quantified. In the ratio of the molar amount of the C-PA protein to the molar amount of the polypeptide of 1:100, the polypeptide solution was slowly added to the precipitates obtained after centrifugation of C-PA-BLP, and mixed slowly. The mixed system was incubated at 2° C. to 8° C. with slow shaking for 48 hours. The obtained product was centrifuged and resuspended with PBS to remove unbound polypeptide.

Example 34: Determination of the Ligation Efficiency of the Phosphorylated Polypeptide Antigen Product Coupled with BLP

(124) A 10 mM dithiothreitol solution was added to the phosphorylated polypeptide antigen product coupled with C-PA-BLP at a volume ratio of 1:1, and mixed homogeneously, and the mixture was left to stand at room temperature to react for 16 hours. The reaction product was centrifuged at 16,000 g at 4° C. for 15 min. The supernatants were taken for HPLC determination. Using 0.5 mg/mL, 0.25 mg/mL, 0.125 mg/mL, 0.0625 mg/mL, 0.03125 mg/mL, 0.03125 mg/mL, and 0.015625 mg/mL phosphorylated polypeptide dissolved in 10 mM of a dithiothreitol solution as standard, a standard curve was plotted with the concentration of the standard versus the peak area of the phosphorylated peptide to quantitatively analyze the concentration of the test sample (the results are shown in FIGS. 13A, 13C and 13D).

(125) It should be understood that specific embodiments described herein are only used for explaining the present invention, instead of limiting the present invention. The protection scope of the present invention is subject to the protection scope defined in claims. For those of ordinary skill in the art, a variety of changes and modifications can be made without departing from the spirit and scope of the present invention, and these changes and modifications should be considered to belong to the protection scope of the present invention.

Claims

1. A phosphorylated polypeptide antigen vaccine, which comprises at least two polypeptide fragments or conservatively modified variants thereof from human full-length Tau protein, wherein the polypeptide fragments or conservatively modified variants thereof contain phosphorylation sites, wherein the polypeptide fragments are connected directly by peptide bonds or connected by amino acid linkers.
2. The phosphorylated polypeptide antigen vaccine of claim 1, wherein the polypeptide fragments are connected directly by peptide bonds.
3. The phosphorylated polypeptide antigen vaccine of claim 1, wherein the polypeptide fragments are derived from phosphorylation modification site-rich regions of human full-length Tau protein.

4. The phosphorylated polypeptide antigen vaccine of claim 3, wherein the phosphorylation sites include two or more of phosphorylated amino acid sites corresponding to positions 18, 202, 205, 212, 214, 231, 235, 238, 262, 396 and 404 of the amino acid sequence of human full-length Tau protein, namely, 18(P-Tyr.sub.18), 202(P-Ser.sub.202), 205(P-Thr.sub.205), 212(P-Thr.sub.212), 214(P-Ser.sub.214), 231(P-Thr.sub.231), 235(P-Ser.sub.235), 238(P-Ser.sub.238), 262(P-Ser.sub.262), 396(P-Ser.sub.396) and 404(P-Ser.sub.404).
5. The phosphorylated polypeptide antigen vaccine of claim 1, wherein the conservatively modified variant of the polypeptide fragment is a variant obtained by conservatively substitution of one or more amino acids of the polypeptide fragment with functionally similar amino acids.
6. The phosphorylated polypeptide antigen vaccine of claim 1, which has an amino acid sequence as represented by any one of SEQ ID NOs: 1-1331 and the amino acid sequence contains two or more phosphorylation sites selected from 18(P-Tyr.sub.18), 202(P-Ser.sub.202), 205(P-Thr.sub.205), 212(P-Thr.sub.212), 214(P-Ser.sub.214), 231(P-Thr.sub.231), 235(P-Ser.sub.235), 238(P-Ser.sub.238), 262(P-Ser.sub.262), 396(P-Ser.sub.396) and 404(P-Ser.sub.404).
7. A complex vaccine formed by coupling the phosphorylated polypeptide antigen vaccine of claim 1 with a carrier.
8. The complex vaccine of claim 7, wherein the carrier is selected from the group consisting of human serum albumin, keyhole limpet hemocyanin, bacterium-like particles (BLP), immunoglobulin molecules, thyroglobulin, ovalbumin, bovine serum albumin component V, influenza hemagglutinin, PAN-DR binding peptide (PADRE polypeptide), malaria circumsporozoite (CS) protein, hepatitis B surface antigen (HB.sub.SAg.sub.19-28), Heat Shock Protein (HSP) 65, Bacille Calmette-Guérin (BCG), cholera toxin, attenuated cholera toxin variants, diphtheria toxin, norovirus capsid P protein, recombinant *Streptococcus* C5a peptidase, *Streptococcus pyogenes* ORF1224, *Streptococcus pyogenes* ORF1664, *Streptococcus pyogenes* ORF2452, pneumolysin, attenuated pneumolysin toxicity variants, *Chlamydia pneumoniae* ORFT367, *Chlamydia pneumoniae* ORFT858, Tetanus toxoid, HIV gp120T1, microbial surface components recognizing adhesive matrix molecules (MSCRAMMS), growth factor/hormone and/or chemokines.
9. A method for preparing the complex vaccine of claim 7, comprising the following steps: 1) artificially synthesizing the phosphorylated polypeptide antigen vaccine of claim 1; 2) preparing a carrier to be coupled to the phosphorylated polypeptide antigen vaccine; 3) mixing the phosphorylated polypeptide antigen vaccine with the carrier to perform coupling reaction; and 4) separating and purifying the conjugate obtained in 3), thereby obtaining a complex vaccine.
10. The method of claim 9, wherein the carrier in step 2) is a norovirus capsid P protein, preferably PP-3C or PP-3K, and step 2) specifically comprises: i) obtaining an expression vector comprising a nucleic acid encoding a PP-3C or PP-3K protein; ii) transferring the expression vector into a receptor cell; iii) expressing the PP-3C or PP-3K protein, and allowing it to self-assemble into a recombinant P particle in the receptor cell; preferably, step 2) also comprises isolation and purification steps; and more preferably, ion exchange chromatography and/or hydrophobic chromatography are used for purification.
11. The method of claim 9, wherein in step 3), PP-3C is used as a vaccine carrier for coupling, a preferred buffer system is an ammonium bicarbonate buffer system, and a preferred pH ranges from 7.5 to 8.8; preferably, the phosphorylated polypeptide antigen vaccine and the carrier are mixed in a ratio of 10:1 to 100:1, and a preferred reaction temperature ranges from 2° C. to 10° C.; or in step 3), PP-3K is used as a vaccine carrier for coupling, and a preferred buffer system is a phosphate buffer system, and a preferred pH ranges from 7.0 to 8.5; preferably, the phosphorylated polypeptide antigen vaccine and the carrier are mixed in a ratio of 10:1 to 100:1, and a preferred reaction temperature ranges from 2° C. to 25° C.
12. The method of claim 9, wherein the carrier in step 2) is bacterium-like particles (BLP), and step 3) specifically comprises: i) obtaining a purified protein adaptor—C-PA protein with a sequence as

represented by SEQ ID NO: 1364; and ii) connecting the carrier—bacterium-like particles (BLP)—obtained in step 2) with the C-PA protein to obtain C-PA-BLP; and iii) performing coupling reaction of C-PA-BLP with the phosphorylated polypeptide antigen vaccine; wherein a Tris buffer system is used as a buffer system and a preferred pH ranges from 7.2 to 8.8; a preferred C-PA protein concentration is 0.1 mg/mL to 1.5 mg/mL, and a preferred reaction temperature ranges from 2° C. to 30° C.; or an ammonium bicarbonate buffer system is used as a buffer system, and a preferred pH ranges from 7.5 to 8.8; preferably, the phosphorylated polypeptide antigen vaccine and the C-PA-BLP are mixed in a ratio of 10:1 to 100:1, and a preferred reaction temperature ranges from 2° C. to 10° C.

13. The method of claim 9, wherein step 4) comprises removing coupling agent and polypeptide antigens which are not successfully connected by methods including desalting chromatography, dialysis and ultrafiltration.

14. A vaccine composition, wherein the vaccine composition comprises the phosphorylated polypeptide antigen vaccine of claim 1 or a complex vaccine; preferably, the vaccine composition further comprises a pharmaceutically acceptable adjuvant; more preferably, the pharmaceutically acceptable adjuvant is selected from one or more of CpG, MF59, AS02, AS03, Freund's complete adjuvant and Freund's incomplete adjuvant; wherein the complex vaccine is formed by coupling the phosphorylated polypeptide antigen vaccine of claim 1 with a carrier.

15. Use of the phosphorylated polypeptide antigen vaccine of claim 1 or a complex vaccine or a vaccine composition for preparing a medicament for prevention and/or treatment of neurodegenerative disorders; and/or for preparing a medicament for maintaining or improving, preferably recovering, and more preferably completely recovering the cognitive memory of mammals, especially human beings; wherein the complex vaccine is formed by coupling the phosphorylated polypeptide antigen vaccine of claim 1 with a carrier; wherein the vaccine composition comprises the phosphorylated polypeptide antigen vaccine of claim 1 or the complex vaccine; preferably, the vaccine composition further comprises a pharmaceutically acceptable adjuvant; more preferably, the pharmaceutically acceptable adjuvant is selected from one or more of CpG, MF59, AS02, AS03, Freund's complete adjuvant and Freund's incomplete adjuvant.

16. Use of claim 15, wherein the neurodegenerative disorders are selected from one or more of AD, Creutzfeldt-Jacob Syndrome, Dementia pugilistica, Down's Syndrome, Gerstmann-Sträussler-Scheinker disease, inclusion-body myositis, prion protein cerebral amyloid angiopathy, traumatic brain injury, amyotrophic lateral sclerosis/parkinsonism syndrome-dementia syndrome, argyrophilic grain dementia, corticobasal degeneration, diffuse neurofibrillary tangles with calcification, frontotemporal dementia with parkinsonism syndrome linked to chromosome 17, Hallevorden-Spatz disease, multiple system atrophy, Niemann-Pick disease type C, Pick's disease, progressive subcortical gliosis and progressive supranuclear panencephalitis; preferably the neurodegenerative disorder is AD; preferably, the vaccine or vaccine composition is preferably immunized subcutaneously, intraperitoneally or intramuscularly, and more preferably immunized intramuscularly.

17. The phosphorylated polypeptide antigen vaccine of claim 1, wherein the phosphorylated polypeptide antigen vaccine contains an additional cysteine residue at its C'-terminal.

18. The phosphorylated polypeptide antigen vaccine of claim 3, wherein the polypeptide fragments are derived from regions of human full-length Tau protein as follows: amino acids at positions 14 to 22 of human full-length Tau protein, amino acids at positions 194 to 266 of human full-length Tau protein, and/or amino acids at positions 392 to 408 of human full-length Tau protein.

19. The phosphorylated polypeptide antigen vaccine of claim 4, wherein the phosphorylation sites include all phosphorylated amino acid sites corresponding to positions 18, 202, 205, 212, 214, 231, 235, 238, 262, 396 and 404 of the amino acid sequence of human full-length Tau protein, namely, 18(P-Tyr.sub.18), 202(P-Ser.sub.202), 205(P-Thr.sub.205), 212(P-Thr.sub.212), 214(P-Ser.sub.214), 231(P-Thr.sub.231), 235(P-Ser.sub.235), 238(P-Ser.sub.238), 262(P-Ser.sub.262),

396(P-Ser.sub.396) and 404(P-Ser.sub.404).

20. The phosphorylated polypeptide antigen vaccine of claim 5, wherein the conservatively modified variant of the polypeptide fragment is a variant obtained by conservatively substitution of 1 to 10 amino acids of the polypeptide fragment with functionally similar amino acids.

21. The phosphorylated polypeptide antigen vaccine of claim 20, wherein the conservatively modified variant of the polypeptide fragment is a variant obtained by conservatively substitution of 1 to 6 amino acids of the polypeptide fragment with functionally similar amino acids.

22. The phosphorylated polypeptide antigen vaccine of claim 21, wherein the conservatively modified variant of the polypeptide fragment is a variant obtained by conservatively substitution of 1 to 4 amino acids of the polypeptide fragment with functionally similar amino acids.

23. The phosphorylated polypeptide antigen vaccine of claim 22, wherein the conservatively modified variant of the polypeptide fragment is a variant obtained by conservatively substitution of 1 to 3 amino acids of the polypeptide fragment with functionally similar amino acids.

24. The phosphorylated polypeptide antigen vaccine of claim 23, wherein the conservatively modified variant of the polypeptide fragment is a variant obtained by conservatively substitution of 1 amino acid of the polypeptide fragment with functionally similar amino acids.

25. The phosphorylated polypeptide antigen vaccine of claim 6, wherein the phosphorylated polypeptide antigen vaccine has an amino acid sequence which has at least 80%, at least 85%, at least 90%, at least 95% sequence identity to any one of SEQ ID NOs: 1-1331.

26. The phosphorylated polypeptide antigen vaccine of claim 25, wherein the phosphorylated polypeptide antigen vaccine has an amino acid sequence which has at least 98% sequence identity to any one of SEQ ID NOs: 1-1331.

27. The phosphorylated polypeptide antigen vaccine of claim 26, wherein the phosphorylated polypeptide antigen vaccine has an amino acid sequence which has at least 99% sequence identity to any one of SEQ ID NOs: 1-1331.

28. The phosphorylated polypeptide antigen vaccine of claim 25, wherein the phosphorylated polypeptide antigen vaccine has an amino acid sequence as represented by any one of SEQ ID NO: 201, SEQ ID NO: 225, SEQ ID NO: 306, SEQ ID NO: 387, SEQ ID NO: 468, SEQ ID NO: 558, SEQ ID NO: 567, SEQ ID NO: 769, SEQ ID NO: 784, SEQ ID NO: 875, SEQ ID NO: 1020, SEQ ID NO: 1101, SEQ ID NO: 1182, SEQ ID NO: 1272, SEQ ID NO: 1313 and SEQ ID NO: 1330, and the amino acid sequence contains two or more phosphorylation sites selected from 18(P-Tyr.sub.18), 202(P-Ser.sub.202), 205(P-Thr.sub.205), 212(P-Thr.sub.212), 214(P-Ser.sub.214), 231(P-Thr.sub.231), 235(P-Ser.sub.235), 238(P-Ser.sub.238), 262(P-Ser.sub.262), 396(P-Ser.sub.396) and 404(P-Ser.sub.404).

29. The phosphorylated polypeptide antigen vaccine of claim 28, wherein the phosphorylated polypeptide antigen vaccine has an amino acid sequence as represented by any one of SEQ ID NO: 201, SEQ ID NO: 225, SEQ ID NO: 306, SEQ ID NO: 387, SEQ ID NO: 468, SEQ ID NO: 558, SEQ ID NO: 567, SEQ ID NO: 769, SEQ ID NO: 784, SEQ ID NO: 875, SEQ ID NO: 1020, SEQ ID NO: 1101, SEQ ID NO: 1182, SEQ ID NO: 1272, SEQ ID NO: 1313 and SEQ ID NO: 1330, and the amino acid sequence respectively contain phosphorylation sites as follows: 18(P-Tyr.sub.18), 202(P-Ser.sub.202), 205(P-Thr.sub.205); 18(P-Tyr.sub.18), 202(P-Ser.sub.202), 205(P-Thr.sub.205); 18(P-Tyr.sub.18), 212(P-Thr.sub.212), 214(P-Ser.sub.214); 18(P-Tyr.sub.18), 231(P-Thr.sub.231), 235(P-Ser.sub.235); 18(P-Tyr.sub.18), 238(P-Ser.sub.238), 262(P-Ser.sub.262); 18(P-Tyr.sub.18), 396(P-Ser.sub.396), 404(P-Ser.sub.404); 202(P-Ser.sub.202), 205(P-Thr.sub.205), 231(P-Thr.sub.231), 235(P-Ser.sub.235); 202(P-Ser.sub.202), 205(P-Thr.sub.205), 231(P-Thr.sub.231), 235(P-Ser.sub.235); 202(P-Ser.sub.202), 205(P-Thr.sub.205), 238(P-Ser.sub.238), 262(P-Ser.sub.262); 202(P-Ser.sub.202), 205(P-Thr.sub.205), 396(P-Ser.sub.396), 404(P-Ser.sub.404); 202(P-Ser.sub.202), 205(P-Thr.sub.205), 396(P-Ser.sub.396), 404(P-Ser.sub.404); 212(P-Thr.sub.212), 214(P-Ser.sub.214), 231(P-Thr.sub.231), 235(P-Ser.sub.235); 212(P-Thr.sub.212), 214(P-Ser.sub.214), 238(P-Ser.sub.238), 262(P-Ser.sub.262); 212(P-Thr.sub.212),

214(P-Ser.sub.214), 396(P-Ser.sub.396), 404(P-Ser.sub.404); 238(P-Ser.sub.238), 262(P-Ser.sub.262), 396(P-Ser.sub.396), 404(P-Ser.sub.404); 238(P-Ser.sub.238), 262(P-Ser.sub.262), 396(P-Ser.sub.396) and 404(P-Ser.sub.404).

30. The complex vaccine of claim 8, wherein the carrier is bacterium-like particles (BLP).

31. The complex vaccine of claim 30, wherein the BLP is coupled with a phosphorylated polypeptide antigen vaccine by means of a protein adaptor.

32. The complex vaccine of claim 31, wherein the protein adaptor has a sequence as represented by SEQ ID NO: 1364.

33. The complex vaccine of claim 8, wherein the carrier is norovirus capsid P protein.

34. The complex vaccine of claim 33, wherein the norovirus capsid P protein is PP-3C with a sequence as represented by SEQ ID NO: 1357 or PP-3K with a sequence as represented by SEQ ID NO: 1359.
